Top 103 CAT Data Interpretation Basics Questions With Video Solutions

Questions

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Details of the top 20 MBA schools in the US as ranked by US News and World Report, 1997 are given below.

School	Overall Ranking	Ranking by Academics	Ranking by Recruiters	Ranking by Placements	Median Starting Salary	% Employed	Annual Tuition Fee
Stanford University	1	1	3	1	$82,000	98.9	$23,100
Harvard University	2	1	2	4	$80,000	96.4	$23,840
University of Pennsylvania	3	1	4	2	$79,000	100.0	$24,956
Massachusetts Institute of Technology	4	1	4	3	$78,000	98.8	$23,900
University of Chicago	5	1	8	10	$65,000	98.4	$23,930
Northwestern University	6	1	1	11	$70,000	93.6	$23,025
Columbia University	7	9	10	5	$83,000	96.2	$23,830
Dartmouth College	8	12	11	6	$70,000	98.3	$23,700
Duke Univrsity	9	9	7	8	$67,500	98.5	$24,380
University of California—Berkeley	10	7	12	12	$70,000	93.7	$18,788
University of Virginia	11	12	9	9	$66,000	98.1	$19,627
University of Michigan—Ann Arbor	12	7	6	14	$65,000	99.1	$23,178
New York University	13	16	19	7	$70,583	97	$23,554
Carnegie Mellon University	14	12	18	13	$67,200	96.6	$22,200
Yale University	15	18	17	22	$65,000	91.5	$23,220
Univ. of North Carolina—Chapel Hill	16	16	16	16	$60,000	96.8	$14,333
University of California—Los Angeles	17	9	13	38	$65,000	82.2	$19,431
University of Texas—Austin	18	18	13	24	$60,000	97.3	$11,614
Indiana University—Bloomington	19	18	20	17	$61,500	95.2	$15,613
Cornell University	20	12	15	36	$64,000	85.1	$23,151

Question 1

Madhu has received admission in all schools listed above. She wishes to select the highest overall ranked school whose annual tuition fee does not exceed $23,000 and median starting salary is at least $70,000. Which school will she select?

A University of Virginia.

B University of Pennsylvania

C Northwestern University

D University of California - Berkeley

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The table below provides certain demographic details of 30 respondents who were part of a survey. The demographic characteristics are: gender, number of children, and age of respondents. The first number in each cell is the number of respondents in that group. The minimum and maximum age of respondents in each group is given in brackets. For example, there are five female respondents with no children and among these five, the youngest is 34 years old, while the oldest is 49.

No. of Children	Male	Female	Total
0	1 (38, 38)	5 (34, 49)	6
1	1 (32, 32)	8 (35, 57)	9
2	8 (21, 65)	3 (37, 63)	11
3	2 (32, 33)	2 (27, 40)	4
Total	12	18	30

Question 2

The percentage of respondents aged less than 40 years is at least

A 10%

B 16.67%

C 20.0%

D 30%

VIEW SOLUTION

Instructions

Answer the questions based on the following information, which gives data about certain coffee producers in India.

	Production in '000 tonnes	Capacity Utilisation (%)	Sales ('000 tonnes)	Total Sales Value (Rs. in crores)
Brooke Bond	2.97	76.5	2.55	31.15
Nestle	2.48	71.2	2.03	26.75
Lipton	1.64	64.8	1.26	15.25
MAC	1.54	59.35	1.47	17.45
Total (Including Others)	11.6	61.3	10.67	132.8

Question 3

What percent of the total market share (by sales value) is controlled by 'others'?

A 60%

B 32%

C 67%

D insufficient data

VIEW SOLUTION

Instructions

In a Class X Board examination, ten papers are distributed over five Groups - PCB, Mathematics, Social Science, Vernacular and English. Each of the ten papers is evaluated out of 100. The final score of a student is calculated in the following manner. First, the Group Scores are obtained by averaging marks in the papers within the Group. The final score is the simple average of the Group Scores. The data for the top ten students are presented below. (Dipan's score in English Paper II has been intentionally removed in the table.)

Name of the Student	PCB Group			Mathematics Group	Social Science Group		Vernacular Group		English Group		Final Score
	Physics	Chemistry	Biology		History	Geography	Paper I	Paper II	Paper I	Paper II	
Ayesha (G)	98	96	97	98	95	93	94	96	96	98	96.2
Ram (B)	97	99	95	97	95	96	94	94	96	98	96.1
Dipan (B)	98	98	98	95	96	95	96	94	96	??	96
Sagnik (B)	97	98	99	96	96	98	94	97	92	94	95.9
Sanjiv (B)	95	96	97	98	97	96	92	93	95	96	95.7
Shreya (G)	96	89	85	100	97	98	94	95	96	95	95.5
Joseph (B)	90	94	98	100	94	97	90	92	94	95	95
Agni (B)	96	99	96	99	95	96	82	93	92	93	94.3
Pritam (B)	98	98	95	98	83	95	90	93	94	94	93.9
Tirna (G)	96	98	97	99	85	94	92	91	87	96	93.7

Note: B or G against the name of a student respectively indicates whether the student is a boy or a girl.

Question 4

Students who obtained Group Scores of at least 95 in every group are eligible to apply for a prize. Among those who are eligible, the student obtaining the highest Group Score in Social Science Group is awarded this prize. The prize was awarded to:

A Shreya

B Ram

C Ayesha

D Dipan

E no one from the top ten

VIEW SOLUTION

Instructions

The table below presents the revenue (in million rupees) of four firms in three states. These firms, Honest Ltd., Aggressive Ltd., Truthful Ltd. and Profitable Ltd. are disguised in the table as A,B,C and D, in no particular order.

States	Firm A	Firm B	Firm C	Firm D
U.P	49	82	80	55
Bihar	69	72	70	65
M.P	72	63	72	65

Further, it is known that:

- In the state of MP, Truthful Ltd. has the highest market share.
- Aggressive Ltd.'s aggregate revenue differs from Honest Ltd.'s by Rs. 5 million.

Question 5

What can be said regarding the following two statements?

Statement 1: Aggressive Ltd.'s lowest revenues are from MP.

Statement 2: Honest Ltd.'s lowest revenues are from Bihar.

A If Statement 2 is true then Statement 1 is necessarily false.

B If Statement 1 is false then Statement 2 is necessarily true.

C If Statement 1 is true then Statement 2 is necessarily true.

D None of the above.

VIEW SOLUTION

Instructions

The table below reports the gender, designation and age-group of the employees in an organization. It also provides information on their commitment to projects coming up in the months of January (Jan), February (Feb), March (Mar) and April (Apr), as well as their interest in attending workshops on: Business Opportunities (BO), Communication Skills (CS), and E-Governance (EG).

Serial No.	Name	Gender	Designation	Age-Group	Committed to Projects During	Interested in Workshop on
1	Anshul	M	Mgr	Y	Jan, Mar	CS, EG
2	Bushkant	M	Dir	I	Feb, Mar	BO, EG
3	Charu	F	Mgr	I	Jan, Feb	BO, CS
4	Dinesh	M	Exe	O	Jan, Apr	BO, CS, EG
5	Eashwaran	M	Dir	O	Feb, Apr	BO
6	Fatima	F	Mgr	Y	Jan, Mar	BO, CS
7	Gayatri	F	Exe	Y	Feb. Mar	EG
8	Hari	M	Mgr	I	Feb, Mar	BO, CS, EG
9	Indira	F	Dir	O	Feb. Apr	BO, EG
10	John	M	Dir	Y	Jan, Mar	BO
11	Kalindi	F	Exe	I	Jan, Apr	BO, CS, EG
12	Lavanya	F	Mgr	O	Feb. Apr	CS, EG
13	Mandeep	M	Mgr	O	Mar, Apr	BO, EG
14	Nandlal	M	Dir	I	Jan, Feb	BO, EG
15	Parul	F	Exe	Y	Feb, Apr	CS,EG
16	Rahul	M	Mgr	Y	Mar, Apr	CS, EG
17	Sunita	F	Dir	Y	Jan, Feb	BO, EG
18	Urvashi	F	Exe	I	Feb, Mar	EG
19	Yamini	F	Mgr	O	Mar, Apr	CS, EG
20	Zeena	F	Exe	Y	Jan, Mar	BO, CS, EG

M=Male, F= Female; Exe=Executive, Mgr=Manager, Dir=Director; Y=Young, I=In-between, O=Old

For each workshop, exactly four employees are to be sent, of which at least two should be Females and at least one should be Young. No employee can be sent to a workshop in which he/she is not interested in. An employee cannot attend the workshop on

- Communication Skills, if he/she is committed to internal projects in the month of January;
- Business Opportunities, if he/she is committed to internal projects in the month of February;
- E-governance, if he/she is committed to internal projects in the month of March.

Question 6

Which set of employees cannot attend any of the workshops?

A Anshul, Charu, Eashwaran and Lavanya

B Anshul, Bushkant, Gayatri, and Urvashi

C Charu, Urvashi, Bushkant and Mandeep

D Anshul, Gayatri, Eashwaran and Mandeep

VIEW SOLUTION

Question 7

How many Executives (Exe) cannot attend more than one workshop?

A 2

B 3

C 15

D 16

VIEW SOLUTION

Instructions

The table below reports annual statistics related to rice production in select states of India for a particular year.

State	Total Area (in millions)	% of Area under rice cultivation	Production (in million tons)	Population (in millions)
Himachal Pradesh	6	20	1.2	6
Kerala	4	60	4.8	32
Rajasthan	34	20	6.8	56
Bihar	10	60	12	83
Karnataka	19	50	19	53
Haryana	4	80	19.2	21
West Bengal	9	80	21.6	80
Gujarat	20	60	24	51
Punjab	5	80	24	24
Madhya Prade3h	31	40	24.8	60
Tamilnadu	13	70	27.3	62
Maharashtra	31	50	48	97
Uttar Pradesh	24	70	67.2	166
Andhra Pradesh	28	80	112	76

Question 8

Which two states account for the highest productivity of rice (tons produced per hectare of rice cultivation)?

A Haryana and Punjab

B Punjab and Andhra Pradesh

C Andhra Pradesh and Haryana

D Uttar Pradesh and Haryana

VIEW SOLUTION

Instructions

Directions for the next 5 questions: Answer these questions based on the data provided in the table below: Factory Sector by Type of Ownership. All figures in the table are in percent of the total for the corresponding column.

Sector	Factories	Employment	Fixed Capital	Gross Output	Value Added
Public	7	27.7	43.2	25.8	30.8
Central Govt.	1	10.5	17.5	12.7	14.1
State/local Govt.	5.2	16.2	24.3	11.6	14.9
Central & State/local Govt.	0.8	1	1.4	1.5	1.8
Joint	1.8	5.1	6.8	8.4	8.1
Wholly private	90.3	64.6	46.8	63.8	58.7
Others	0.9	2.6	3.2	2	2.4
Total	100	100	100	100	100

Question 9

Among the firms in different sectors, value added per employee is highest in:

A Central government

B Central and State/local governments

C Joint sector

D Wholly private

Question 10

Suppose the average employment level is 60 per factory. The average employment in wholly private” factories is approximately:

A 43

B 47

C 50

D 54

Instructions

Directions for following three questions: Answer the following questions based on the information given below: There are 100 employees in an organization across five departments. The following table gives the department-wise distribution of average age, average basic pay and allowances. The gross pay of an employee is the sum of his/her basic pay and allowances.

There are limited numbers of employees considered for transfer/promotion across departments Whenever a person is transferred/promoted from a department of lower average age to a department of higher average age, he/she will get an additional allowance of 10% of basic pay over and above his/her current allowance. There will not be any change in pay structure if a person is transferred/promoted from a department with higher average age to a department with lower average age.

Department	Number of Employees	Average Age	Average Basic Pay	Allowances (% of Basic Pay)
HR	5	45	5000	70
Marketing	30	35	6000	80
Finance	20	30	6500	60
Business Development	35	42	7500	75
Maintenance	10	35	5500	50

Questions below are independent of each other.

Question 11

What is the approximate percentage change in the average gross of the HR department due to transfer of a 40-year old person with basic pay of Rs. 8000 from the Marketing department?

A 9%

B 11%

C 13%

D 15%

E 17%

VIEW SOLUTION

Instructions

Answer the questions based on the following information, which gives data about certain coffee producers in India.

	Production in '000 tonnes	Capacity Utilisation (%)	Sales ('000 tonnes)	Total Sales Value (Rs. in crores)
Brooke Bond	2.97	76.5	2.55	31.15
Nestle	2.48	71.2	2.03	26.75
Lipton	1.64	64.8	1.26	15.25
MAC	1.54	59.35	1.47	17.45
Total (Including Others)	11.6	61.3	10.67	132.8

Question 12

What is the maximum production capacity (in '000 tonnes) of Lipton for coffee?

A 2.53

B 2.85

C 2.24

D 2.07

VIEW SOLUTION

Instructions

Answer the questions based on the following information.

Krishna distributed 10-acre land to Gopal and Ram who paid him the total amount in the ratio 2 : 3. Gopal invested a further Rs. 2 lakh in the land and planted coconut and lemon trees in the ratio 5 : 1 on equal areas of land. There were a total of 100 lemon trees. The cost of one coconut was Rs. 5. The crop took 7 years to mature and when the crop was reaped in 1997, the total revenue generated was 25% of the total amount put in by Gopal and Ram together. The revenue generated from the coconut and lemon trees was in the ratio 3 : 2 and it was shared equally by Gopal and Ram as the initial amount spent by them were equal.

Question 13

What was the total output of coconuts?

A 24,000

B 36,000

C 18,000

D 48,000

VIEW SOLUTION

Question 14

What was the value of output per acre of lemon trees planted?

A 0.24 lakh per acre

B 2.4 lakh per acre

C 24 lakh per acre

D Cannot be determined

VIEW SOLUTION

Instructions

Use the following information: Prakash has to decide whether or not to test a batch of 1000 widgets before sending them to the buyer.
In case he decides to test, he has two options:
a) Use test I ;
b) Use test II.

Test I cost Rs. 2 per widget. However, the test is not perfect. It sends 20% of the bad ones to the buyer as good. Test II costs Rs. 3 per widget. It brings out all the bad ones. A defective widget identified before sending can be corrected at a cost of Rs. 25 per widget. All defective widgets are identified at the buyer's end and penalty of Rs. 50 per defective widget has to be paid by Prakash.

Question 15

If Prakash is told that the lot has 160 defective widgets, he should:

A use Test I only

B use Test II only.

C do no testing.

D either use Test I or do not test.

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions based on the table given below concerning the busiest 20 international airports in the world.

No.	Name	International Airport Type	Code	Location	Passengers
1	Hartsfield	A	ATL	Atlanta, Georgia, USA	77,939,536
2	Chicago-O'Hare	A	ORD	Chicago, Illinois, USA	72,568,076
3	Los Angeles	A	LAX	Los Angeles, California, USA	63,876,561
4	Heathrow	E	LHR	London, United Kingdom	62,263,710
5	DFW	A	DFW	Dallas/Ft. Worth, Texas, USA	60,000,125
6	Haneda Airport	F	HND	Tokyo, Japan	54,338,212
7	Frankfurt Airport	E	FRA	Frankfurt, Germany	45,858,315
8	Roissy-Charles de Gaulle	E	CDG	Paris, France	43,596,943
9	San Francisco	A	SFO	San Francisco, California, USA	40,387,422
10	Denver	A	DIA	Denver, Colorado, USA	38,034,231
11	Amsterdam Schiphol	E	AMS	Amsterdam, Netherlands	36,781,015
12	Minneapolis - St. Paul	A	MSP	Minneapolis-St. Paul, USA	34,216,331
13	Detroit Metropolitan	A	DTW	Detroit, Michigan, USA	34,038,381
14	Miami	A	MIA	Miami, Florida, USA	33,899,246
15	Newark	A	EWR	Newark, New Jersey, USA	33,814,000
16	McCarran	A	LAS	Las Vegas, Nevada, USA	33,669,185
17	Phoenix Sky Harbor	A	PHX	Phoenix, Arizona, USA	33,533,353
18	Kimpo	FE	SEL	Seoul, Korea	33,371,074
19	George Bush	A	IAH	Houston, Texas, USA	33,089,333
20	John F. Kennedy	A	JFK	New York, New York, USA	32,003,000

Question 16

What percentage of top ten busiest airports is in the United States of America?

A 60%

B 80%

C 70%

D 90%

VIEW SOLUTION

Question 17

Of the five busiest airports, roughly, what percentage of passengers in handled by Heathrow Airport?

A 30

B 40

C 20

D 50

VIEW SOLUTION

Instructions

The following table gives the quantities of apples (in tonnes) arriving at New Delhi market from various states in a particular year. The month in which demand was more than supply, the additional demand was met by the stock from cold storage.

Month	State			Cold Storage	Total
	HP	UP	J&K		
April	7	0	7	59	73
May	12	1	0	0	13
June	9,741	257	8,017	0	18,015
July	71,497	0	18,750	0	90,247
August	77,675	0	20,286	0	97,961
September	53,912	0	56,602	0	110,514
October	12,604	0	79,591	24	92,219
November	3,499	0	41,872	42	45,413
December	1,741	0	14,822	15	16,578
January	315	0	10,922	201	11,438
February	25	0	11,183	77	11,285
March	0	0	683	86	769

Question 18

What was the maximum percentage of apples supplied by any state in any of the months?

A 99%

B 95%

C 88%

D 100%

VIEW SOLUTION

Instructions

Read the following information and answer the questions that follows: Ghosh Babu deposited a certain sum of money in a bank in 1986. The bank calculated interest on the principal at 10 percent simple interest, and credited it to the account once a year. After the 1st year, Ghosh Babu withdrew the entire interest and 20% of the initial amount. After the 2nd year, he withdrew the interest and 50% of the remaining amount. After the 3rd year, he withdrew the interest and 50% of the remaining amount. Finally after the 4th year, Ghosh Babu closed the account and collected the entire balance of Rs. 11,000.

Question 19

The year, at the end of which, Ghosh Babu withdrew the maximum amount was:

A First

B Second

C Third

D Fourth

VIEW SOLUTION

Instructions

DIRECTIONS for the following four questions:

A low-cost airline company connects ten India cities, A to J. The table below gives the distance between a pair of airports and the corresponding price charged by the company. Travel is permitted only from a departure airport to an arrival airport. The customers do not travel by a route where they have to stop at more than two intermediate airports.

Sector No.	Airport of Departure	Airport of Arrival	Distance between the airports	Price (Rs.)
1	A	B	560	670
2	A	C	790	1350
3	A	D	850	1250
4	A	E	1245	1600
5	A	F	1345	1700
6	A	G	1350	2450
7	A	H	1950	1850
8	B	C	1650	2000
9	B	H	1750	1900
10	B	I	2100	2450
11	B	J	2300	2275
12	C	D	460	450
13	C	F	410	430
14	C	G	910	1100
15	D	E	540	590
16	D	F	625	700
17	D	G	640	750
18	D	H	950	1250
19	D	J	1650	2450
20	E	F	1250	1700
21	E	G	970	1150
22	E	H	850	875
23	F	G	900	1050
24	F	I	875	950
25	F	J	970	1150
26	G	I	510	550
27	G	J	830	890
28	H	I	790	970
29	H	J	400	425
30	I	J	460	540

Question 20

If the prices include a margin of 15% over the total cost that the company incurs, which among the following is the distance to be covered in flying from A to J that minimizes the total cost of travel for the company?

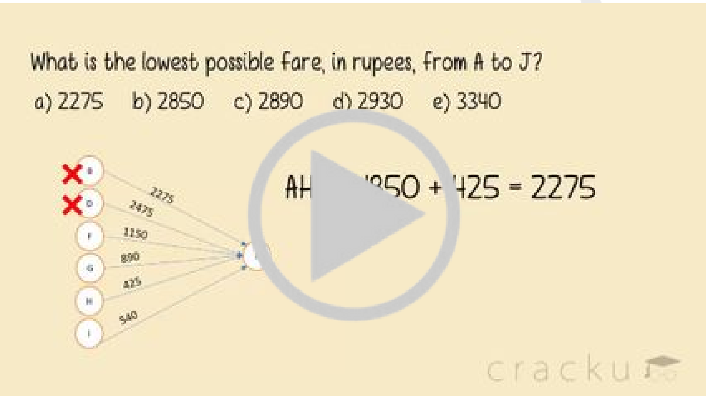
A 2170

B 2180

C 2315

D 2350

E 2390



VIDEO SOLUTION

Instructions

In a Class X Board examination, ten papers are distributed over five Groups - PCB, Mathematics, Social Science, Vernacular and English. Each of the ten papers is evaluated out of 100. The final score of a student is calculated in the following manner. First, the Group Scores are obtained by averaging marks in the papers within the Group. The final score is the simple average of the Group Scores. The data for the top ten students are presented below. (Dipan's score in English Paper II has been intentionally removed in the table.)

Name of the Student	PCB Group			Mathematics Group	Social Science Group		Vernacular Group		English Group		Final Score
	Physics	Chemistry	Biology		History	Geography	Paper I	Paper II	Paper I	Paper II	
Ayesha (G)	98	96	97	98	95	93	94	96	96	98	96.2
Ram (B)	97	99	95	97	95	96	94	94	96	98	96.1
Dipan (B)	98	98	98	95	96	95	96	94	96	??	96
Sagnik (B)	97	98	99	96	96	98	94	97	92	94	95.9
Sanjiv (B)	95	96	97	98	97	96	92	93	95	96	95.7
Shreya (G)	96	89	85	100	97	98	94	95	96	95	95.5
Joseph (B)	90	94	98	100	94	97	90	92	94	95	95
Agni (B)	96	99	96	99	95	96	82	93	92	93	94.3
Pritam (B)	98	98	95	98	83	95	90	93	94	94	93.9
Tirna (G)	96	98	97	99	85	94	92	91	87	96	93.7

Note: B or G against the name of a student respectively indicates whether the student is a boy or a girl.

Question 21

Among the top ten students, how many boys scored at least 95 in at least one paper from each of the groups?

A 1

B 2

C 3

D 4

E 5

VIEW SOLUTION

Instructions

The following table shows the top 20 busiest airports (in terms of passengers) in India. The Passengers column gives the number of passengers handled by each airport in a year.

No.	Name	Airport Rating	Code	Location	Passengers (in one year)
1	Rajiv Gandhi Airport	1	RGIA	Hyderabad, AP	7889987
2	Sardar Patel Airport	2	SVIA	Ahmedabad, Gujarat	7654321
3	Dibrugarh Airport	1	DGA	Dibrugarh, Assam	6890445
4	Gaya International Airport	1	GIA	Gaya, Bihar	6178901
5	Vskp International Airport	3	VIA	Vizag, AP	6000897
6	Shivaji Intnl Airport	1	CSIA	Mumbai, Maharashtra	5897561
7	Zero Airport	3	ZEA	Ziro, Arunachal Pradesh	4567892
8	Chandigarh Airport	1	CA	Chandigarh, Punjab	4237891
9	Vijayawada Airport	3	VZA	Vijayawada, AP	4145673
10	Purnea Int Airport	2	PIA	Purnea, Bihar	3898767
11	Indira Gandhi Airport	1	IGIA	New Delhi, Delhi	3587901
12	Hisar Airport	2	HA	Hisar, Haryana	3567879
13	Rajamundry Airport	1	RJA	Rajamundry, AP	3398764
14	Mehsana Airport	2	MA	Mehsana, Gujarat	3223786
15	Nagarjuna Sagar Airport	3	NSA	Karnool, AP	3217865
16	Bhuntar Airport	1	BIA	Kullu, Himachal	3134761
17	Kannur Intnl Airport	3	KIA	Kannur, Kerala	3113456
18	Sri Sathya Sai Airport	1	SSSA	Putaparthy, AP	3078934
19	Tirupati Intnl Airport	2	TIA	Tirupati, AP	2987434
20	HAL Airport	1	HBA	Bangalore, Karnataka	2923985

Question 22

How many airports of type '1' handle more than 35 lakh passengers each?

VIEW SOLUTION

Instructions

Read the following information and answer the questions that follows: Ghosh Babu deposited a certain sum of money in a bank in 1986. The bank calculated interest on the principal at 10 percent simple interest, and credited it to the account once a year. After the 1st year, Ghosh Babu withdrew the entire interest and 20% of the initial amount. After the 2nd year, he withdrew the interest and 50% of the remaining amount. After the 3rd year, he withdrew the interest and 50% of the remaining amount. Finally after the 4th year, Ghosh Babu closed the account and collected the entire balance of Rs. 11,000.

Question 23

The initial amount in rupees, deposited by Ghosh Babu was:

A 25,000

B 75,000

C 50,000

D None of these

VIEW SOLUTION

Instructions

In a certain board examination, students were to appear for examination in five subjects:
English, Hindi, Mathematics, Science and Social Science. Due to a certain emergency situation, a few of the examinations could not be conducted for some students. Hence, some students missed one examination and some others missed two examinations. Nobody missed more than two examinations.

The board adopted the following policy for awarding marks to students. If a student appeared in all five examinations, then the marks awarded in each of the examinations were on the basis of the scores obtained by them in those examinations.

If a student missed only one examination, then the marks awarded in that examination was the average of the best three among the four scores in the examinations they appeared for. If a student missed two examinations, then the marks awarded in each of these examinations was the average of the best two among the three scores in the examinations they appeared for. The marks obtained by six students in the examination are given in the table below. Each of them missed either one or two examinations.

	English	Hindi	Mathematics	Science	Social Science
Alva	80	75	70	75	60
Bithi	90	80	55	85	85
Carl	75	80	90	100	90
Deep	70	90	100	90	80
Esha	80	85	95	60	55
Foni	83	72	78	88	83

The following facts are also known.
I. Four of these students appeared in each of the English, Hindi, Science, and Social Science examinations.
II. The student who missed the Mathematics examination did not miss any other examination.
III. One of the students who missed the Hindi examination did not miss any other examination. The other student who missed the Hindi examination also missed the Science examination.

Question 24

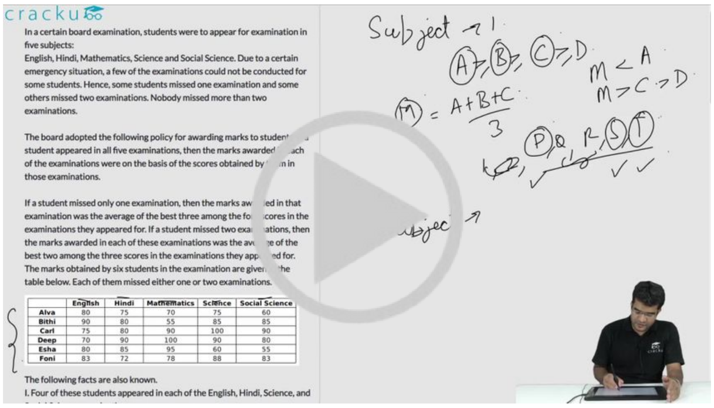
Which students did not appear for the English examination?

A Carl and Deep

B Cannot be determined

C Alva and Bithi

D Esha and Foni



VIDEO SOLUTION

Top-500 Free CAT Questions (With Solutions)

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The table below provides certain demographic details of 30 respondents who were part of a survey. The demographic characteristics are: gender, number of children, and age of respondents. The first number in each cell is the number of respondents in that group. The minimum and maximum age of respondents in each group is given in brackets. For example, there are five female respondents with no children and among these five, the youngest is 34 years old, while the oldest is 49.

No. of Children	Male	Female	Total
0	1 (38, 38)	5 (34, 49)	6
1	1 (32, 32)	8 (35, 57)	9
2	8 (21, 65)	3 (37, 63)	11
3	2 (32, 33)	2 (27, 40)	4
Total	12	18	30

Question 25

The percentage of respondents that fall into the 35 to 40 years age group (both inclusive) is at least

A 6.67%

B 10%

C 13.33%

D 26.67%

VIEW SOLUTION

Instructions

Directions for following three questions: Answer the following questions based on the information given below: There are 100 employees in an organization across five departments. The following table gives the department-wise distribution of average age, average basic pay and allowances. The gross pay of an employee is the sum of his/her basic pay and allowances.

There are limited numbers of employees considered for transfer/promotion across departments Whenever a person is transferred/promoted from a department of lower average age to a department of higher average age, he/she will get an additional allowance of 10% of basic pay over and above his/her current allowance. There will not be any change in pay structure if a person is transferred/promoted from a department with higher average age to a department with lower average age.

Department	Number of Employees	Average Age	Average Basic Pay	Allowances (% of Basic Pay)
HR	5	45	5000	70
Marketing	30	35	6000	80
Finance	20	30	6500	60
Business Development	35	42	7500	75
Maintenance	10	35	5500	50

Questions below are independent of each other.

Question 26

There was a mutual transfer of an employee between Marketing and Finance departments and transfer of one employee from Marketing to HR. As a result, the average age of finance department increased by one year and that of Marketing department remained the same. What is the new average age of HR department?

A 30

B 35

C 40

D 45

E cannot be determined

VIEW SOLUTION

Instructions

In a certain board examination, students were to appear for examination in five subjects:
English, Hindi, Mathematics, Science and Social Science. Due to a certain emergency situation, a few of the examinations could not be conducted for some students. Hence, some students missed one examination and some others missed two examinations. Nobody missed more than two examinations.

The board adopted the following policy for awarding marks to students. If a student appeared in all five examinations, then the marks awarded in each of the examinations were on the basis of the scores obtained by them in those examinations.

If a student missed only one examination, then the marks awarded in that examination was the average of the best three among the four scores in the examinations they appeared for. If a student missed two examinations, then the marks awarded in each of these examinations was the average of the best two among the three scores in the examinations they appeared for. The marks obtained by six students in the examination are given in the table below. Each of them missed either one or two examinations.

	English	Hindi	Mathematics	Science	Social Science
Alva	80	75	70	75	60
Bithi	90	80	55	85	85
Carl	75	80	90	100	90
Deep	70	90	100	90	80
Esha	80	85	95	60	55
Foni	83	72	78	88	83

The following facts are also known.
I. Four of these students appeared in each of the English, Hindi, Science, and Social Science examinations.
II. The student who missed the Mathematics examination did not miss any other examination.
III. One of the students who missed the Hindi examination did not miss any other examination. The other student who missed the Hindi examination also missed the Science examination.

Question 27

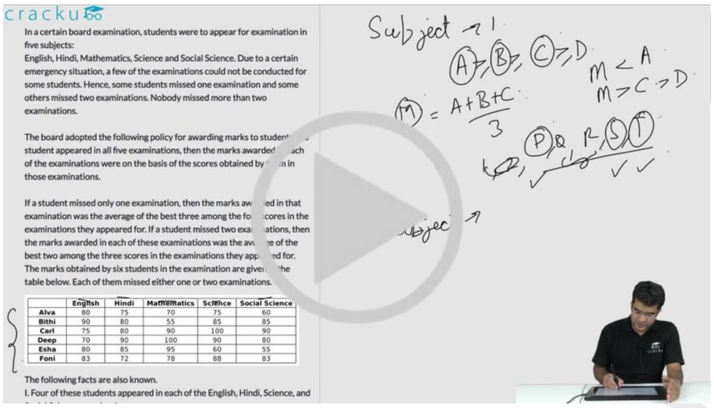
What BEST can be concluded about the students who did not appear for the Hindi examination?

A Deep and Esha

B Alva and Deep

C Alva and Esha

D Two among Alva, Deep and Esha



VIDEO SOLUTION

Free CAT Study Material

Instructions

The table below reports annual statistics related to rice production in select states of India for a particular year.

State	Total Area (in millions)	% of Area under rice cultivation	Production (in million tons)	Population (in millions)
Himachal Pradesh	6	20	1.2	6
Kerala	4	60	4.8	32
Rajasthan	34	20	6.8	56
Bihar	10	60	12	83
Karnataka	19	50	19	53
Haryana	4	80	19.2	21
West Bengal	9	80	21.6	80
Gujarat	20	60	24	51
Punjab	5	80	24	24
Madhya Prade3h	31	40	24.8	60
Tamilnadu	13	70	27.3	62
Maharashtra	31	50	48	97
Uttar Pradesh	24	70	67.2	166
Andhra Pradesh	28	80	112	76

Question 28

An intensive rice producing state is defined as one whose annual rice production per million of population is at least 400,000 tons. How many states are intensive rice producing states?

A 5

B 6

C 7

D 8

VIEW SOLUTION

Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

Figures in Rs.	A	B	C	D	Total
Sales	24,568	25,468	23,752	15,782	89,570
Operating Costs	17,198	19,101	16,151	10,258	62,708
Interest Costs	2,457	2,292	2,850	1,578	9,177
Profit	4,914	4,075	4,750	3,946	17,685

An industry comprises four firms (A, B, C, and D). Financial details of these firms and of the industry as a whole for a particular year are given below. Profitability of a firm is defined as profit as a percentage of sales.

Question 29

If firm A acquires firm B, approximately what percentage of the total market (total sales) will they corner together?

A 55%

B 45%

C 35%

D 50%

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

A study was conduced to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

	Country				
Rank	India	China	Japan	Malaysia	Thailand
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

Question 30

Which amongst the following countries is most dissimilar to India?

A China

B Japan

C Malaysia

D Thailand

VIEW SOLUTION

Free CAT Preparation Video Lectures

Instructions

Directions for the next 5 questions: Answer these questions based on the data provided in the table below: Factory Sector by Type of Ownership. All figures in the table are in percent of the total for the corresponding column.

Sector	Factories	Employment	Fixed Capital	Gross Output	Value Added
Public	7	27.7	43.2	25.8	30.8
Central Govt.	1	10.5	17.5	12.7	14.1
State/local Govt.	5.2	16.2	24.3	11.6	14.9
Central & State/local Govt.	0.8	1	1.4	1.5	1.8
Joint	1.8	5.1	6.8	8.4	8.1
Wholly private	90.3	64.6	46.8	63.8	58.7
Others	0.9	2.6	3.2	2	2.4
Total	100	100	100	100	100

Question 31

Capital productivity is defined as the gross output value per rupee of fixed capital. The three sectors with the higher capital productivity, arranged in descending order are:

A Joint, wholly private, central and state/local

B Wholly private, joint, central and state/local

C Wholly private, central and state/local, joint

D joint, wholly private, central.

VIEW SOLUTION

Instructions

XYZ organization got into the business of delivering groceries to home at the beginning of the last month. They have a two-day delivery promise. However, their deliveries are unreliable. An order booked on a particular day may be delivered the next day or the day after. If the order is not delivered at the end of two days, then the order is declared as lost at the end of the second day. XYZ then does not deliver the order, but informs the customer, marks the order as lost, returns the payment and pays a penalty for non-delivery. The following table provides details about the operations of XYZ for a week of the last month. The first column gives the date, the second gives the cumulative number of orders that were booked up to and including that day. The third column represents the number of orders delivered on that day. The last column gives the cumulative number of orders that were lost up to and including that day. It is known that the numbers of orders that were booked on the 11th, 12th, and 13th of the last month that took two days to deliver were 4, 6, and 8 respectively

Day	Cumilative orders booked	Orders delivered on day	Cumilative orders lost
13th	219	11	91
14th	249	27	92
15th	277	23	94
16th	302	11	106
17th	327	21	118
18th	332	13	120
19th	337	14	129

Question 32

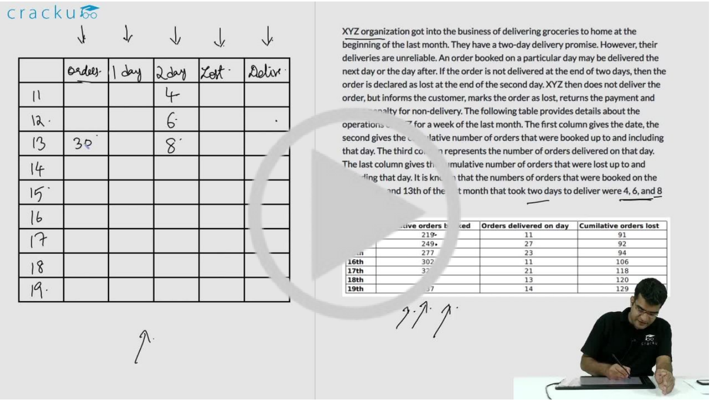
Among the following days, the largest fraction of orders booked on which day was lost?

A 15th

B 16th

C 13th

D 14th



VIDEO SOLUTION

Instructions

Answer the questions based on the following information, which gives data about certain coffee producers in India.

	Production in '000 tonnes	Capacity Utilisation (%)	Sales ('000 tonnes)	Total Sales Value (Rs. in crores)
Brooke Bond	2.97	76.5	2.55	31.15
Nestle	2.48	71.2	2.03	26.75
Lipton	1.64	64.8	1.26	15.25
MAC	1.54	59.35	1.47	17.45
Total (Including Others)	11.6	61.3	10.67	132.8

Question 33

What is the approximate total production capacity (in ‘000 tonnes) for coffee in India?

A 18

B 20

C 18.9

D Data insufficient

VIEW SOLUTION

Instructions

XYZ organization got into the business of delivering groceries to home at the beginning of the last month. They have a two-day delivery promise. However, their deliveries are unreliable. An order booked on a particular day may be delivered the next day or the day after. If the order is not delivered at the end of two days, then the order is declared as lost at the end of the second day. XYZ then does not deliver the order, but informs the customer, marks the order as lost, returns the payment and pays a penalty for non-delivery. The following table provides details about the operations of XYZ for a week of the last month. The first column gives the date, the second gives the cumulative number of orders that were booked up to and including that day. The third column represents the number of orders delivered on that day. The last column gives the cumulative number of orders that were lost up to and including that day. It is known that the numbers of orders that were booked on the 11th, 12th, and 13th of the last month that took two days to deliver were 4, 6, and 8 respectively

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14th	249	27	92
15th	277	23	94
16th	302	11	106
17th	327	21	118
18th	332	13	120
19th	337	14	129

Question 34

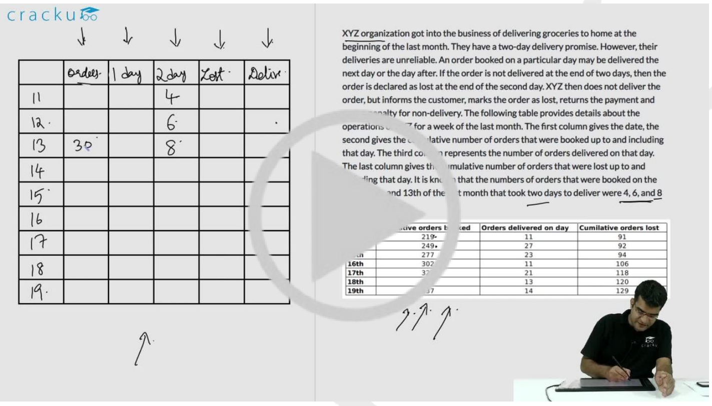
The average time taken to deliver orders booked on a particular day is computed as follows. Let the number of orders delivered the next day be x and the number of orders delivered the day after be y. Then the average time to deliver order is $\frac{(x+2y)}{(x+y)}$. On which of the following days was the average time taken to deliver orders booked the least?

A 15th

B 13th

C 14th

D 16th



VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

A study was conduced to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

	Country				
Rank	India	China	Japan	Malaysia	Thailand
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

Question 35

Which of the following pairs of countries are most dissimilar?

A China and Japan

B India and China

C Malaysia and Japan

D Thailand and Japan

VIEW SOLUTION

Instructions

Read the following information and answer the questions that follows: Ghosh Babu deposited a certain sum of money in a bank in 1986. The bank calculated interest on the principal at 10 percent simple interest, and credited it to the account once a year. After the 1st year, Ghosh Babu withdrew the entire interest and 20% of the initial amount. After the 2nd year, he withdrew the interest and 50% of the remaining amount. After the 3rd year, he withdrew the interest and 50% of the remaining amount. Finally after the 4th year, Ghosh Babu closed the account and collected the entire balance of Rs. 11,000.

Question 36

The year, at the end of which, Ghosh Babu collected the maximum interest was:

A First

B Second

C Third

D Fourth

VIEW SOLUTION

Instructions

XYZ organization got into the business of delivering groceries to home at the beginning of the last month. They have a two-day delivery promise. However, their deliveries are unreliable. An order booked on a particular day may be delivered the next day or the day after. If the order is not delivered at the end of two days, then the order is declared as lost at the end of the second day. XYZ then does not deliver the order, but informs the customer, marks the order as lost, returns the payment and pays a penalty for non-delivery. The following table provides details about the operations of XYZ for a week of the last month. The first column gives the date, the second gives the cumulative number of orders that were booked up to and including that day. The third column represents the number of orders delivered on that day. The last column gives the cumulative number of orders that were lost up to and including that day. It is known that the numbers of orders that were booked on the 11th, 12th, and 13th of the last month that took two days to deliver were 4, 6, and 8 respectively

Day	Cumilative orders booked	Orders delivered on day	Cumilative orders lost
13th	219	11	91
14th	249	27	92
15th	277	23	94
16th	302	11	106
17th	327	21	118
18th	332	13	120
19th	337	14	129

Question 37

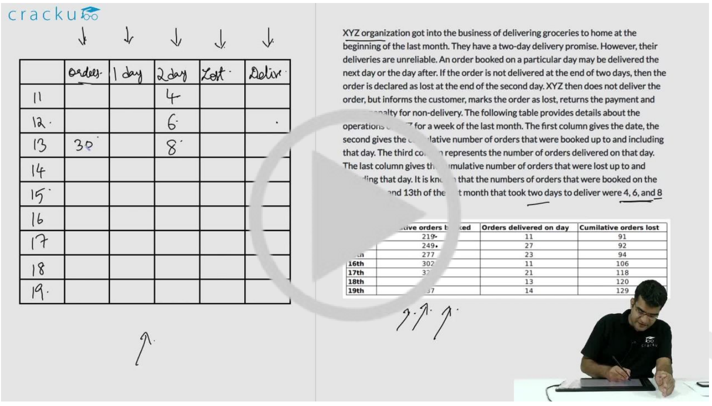
On which of the following days was the number of orders booked the highest?

A 12th

B 15th

C 13th

D 14th



VIDEO SOLUTION

Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

Figures in Rs.	A	B	C	D	Total
Sales	24,568	25,468	23,752	15,782	89,570
Operating Costs	17,198	19,101	16,151	10,258	62,708
Interest Costs	2,457	2,292	2,850	1,578	9,177
Profit	4,914	4,075	4,750	3,946	17,685

An industry comprises four firms (A, B, C, and D). Financial details of these firms and of the industry as a whole for a particular year are given below. Profitability of a firm is defined as profit as a percentage of sales.

Question 38

Which firm has the highest profitability?

A A

B B

C C

D D

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The table below provides certain demographic details of 30 respondents who were part of a survey. The demographic characteristics are: gender, number of children, and age of respondents. The first number in each cell is the number of respondents in that group. The minimum and maximum age of respondents in each group is given in brackets. For example, there are five female respondents with no children and among these five, the youngest is 34 years old, while the oldest is 49.

No. of Children	Male	Female	Total
0	1 (38, 38)	5 (34, 49)	6
1	1 (32, 32)	8 (35, 57)	9
2	8 (21, 65)	3 (37, 63)	11
3	2 (32, 33)	2 (27, 40)	4
Total	12	18	30

Question 39

Given the information above, the percentage of respondents older than 35 can be at most

A 30%

B 73.33%

C 76.67%

D 90%

VIEW SOLUTION

CAT Percentile Predictor

Instructions

Directions for the next 5 questions: Answer these questions based on the data provided in the table below: Factory Sector by Type of Ownership. All figures in the table are in percent of the total for the corresponding column.

Sector	Factories	Employment	Fixed Capital	Gross Output	Value Added
Public	7	27.7	43.2	25.8	30.8
Central Govt.	1	10.5	17.5	12.7	14.1
State/local Govt.	5.2	16.2	24.3	11.6	14.9
Central & State/local Govt.	0.8	1	1.4	1.5	1.8
Joint	1.8	5.1	6.8	8.4	8.1
Wholly private	90.3	64.6	46.8	63.8	58.7
Others	0.9	2.6	3.2	2	2.4
Total	100	100	100	100	100

Question 40

The total value added in all sectors is estimated at Rs. 14,000 crores. Suppose that the number of firms in the joint sector is 2700. The average value added per factory, in Rs. crores, in the central govt. is:

A 1.41

B 14.1

C 13.2

D 1.32

VIEW SOLUTION

Instructions

These questions are based on the table below presenting data on percentage population covered by drinking water and sanitation facilities in selected Asian countries.

	Drinking water			Sanitation facilities		
	Urban	Rural	Total	Urban	Rural	Total
India	85	79	81	70	14	29
Bangladesh	99	96	97	79	44	48
China	97	56	67	74	7	24
Pakistan	82	69	74	77	22	47
Philippines	92	80	86	88	66	77
Indonesia	79	54	62	73	40	51
Sri Lanka	88	52	57	68	62	63
Nepal	88	60	63	58	12	1

(*Source:* World Resources 1998-99, p. 251, UNDP, UNEP and World Bank.)

Country A is said to dominate B or A > B if A has higher percentage in total coverage for both drinking water and sanitation facilities, and, B is said to be dominated by A, or B < A. A country is said to be on the coverage frontier if no other country dominates it. Similarly, a country is not on the coverage frontier if it is dominated by at least one other country.

Question 41

Again, using only the data presented under Sanitation facilities columns, sequence China, Indonesia and Philippines in ascending order of rural population as a percentage of their respective total populations. The correct order is:

A Philippines, Indonesia, China

B Indonesia, China, Philippines

C Indonesia, Philippines, China

D China, Indonesia, Philippines

VIEW SOLUTION

Instructions

In a Class X Board examination, ten papers are distributed over five Groups - PCB, Mathematics, Social Science, Vernacular and English. Each of the ten papers is evaluated out of 100. The final score of a student is calculated in the following manner. First, the Group Scores are obtained by averaging marks in the papers within the Group. The final score is the simple average of the Group Scores. The data for the top ten students are presented below. (Dipan's score in English Paper II has been intentionally removed in the table.)

Name of the Student	PCB Group			Mathematics Group	Social Science Group		Vernacular Group		English Group		Final Score
	Physics	Chemistry	Biology		History	Geography	Paper I	Paper II	Paper I	Paper II	
Ayesha (G)	98	96	97	98	95	93	94	96	96	98	96.2
Ram (B)	97	99	95	97	95	96	94	94	96	98	96.1
Dipan (B)	98	98	98	95	96	95	96	94	96	??	96
Sagnik (B)	97	98	99	96	96	98	94	97	92	94	95.9
Sanjiv (B)	95	96	97	98	97	96	92	93	95	96	95.7
Shreya (G)	96	89	85	100	97	98	94	95	96	95	95.5
Joseph (B)	90	94	98	100	94	97	90	92	94	95	95
Agni (B)	96	99	96	99	95	96	82	93	92	93	94.3
Pritam (B)	98	98	95	98	83	95	90	93	94	94	93.9
Tirna (G)	96	98	97	99	85	94	92	91	87	96	93.7

Note: B or G against the name of a student respectively indicates whether the student is a boy or a girl.

Question 42

Each of the ten students was allowed to improve his/her score in exactly one paper of choice with the objective of maximizing his /her final score. Everyone scored 100 in the paper in which he or she chose to improve. After that, the topper among the ten students was:

A Ram

B Agni

C Pritam

D Ayesha

E Dipan

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Details of the top 20 MBA schools in the US as ranked by US News and World Report, 1997 are given below.

School	Overall Ranking	Ranking by Academics	Ranking by Recruiters	Ranking by Placements	Median Starting Salary	% Employed	Annual Tuition Fee
Stanford University	1	1	3	1	$82,000	98.9	$23,100
Harvard University	2	1	2	4	$80,000	96.4	$23,840
University of Pennsylvania	3	1	4	2	$79,000	100.0	$24,956
Massachusetts Institute of Technology	4	1	4	3	$78,000	98.8	$23,900
University of Chicago	5	1	8	10	$65,000	98.4	$23,930
Northwestern University	6	1	1	11	$70,000	93.6	$23,025
Columbia University	7	9	10	5	$83,000	96.2	$23,830
Dartmouth College	8	12	11	6	$70,000	98.3	$23,700
Duke Univrsity	9	9	7	8	$67,500	98.5	$24,380
University of California—Berkeley	10	7	12	12	$70,000	93.7	$18,788
University of Virginia	11	12	9	9	$66,000	98.1	$19,627
University of Michigan—Ann Arbor	12	7	6	14	$65,000	99.1	$23,178
New York University	13	16	19	7	$70,583	97	$23,554
Carnegie Mellon University	14	12	18	13	$67,200	96.6	$22,200
Yale University	15	18	17	22	$65,000	91.5	$23,220
Univ. of North Carolina—Chapel Hill	16	16	16	16	$60,000	96.8	$14,333
University of California—Los Angeles	17	9	13	38	$65,000	82.2	$19,431
University of Texas—Austin	18	18	13	24	$60,000	97.3	$11,614
Indiana University—Bloomington	19	18	20	17	$61,500	95.2	$15,613
Cornell University	20	12	15	36	$64,000	85.1	$23,151

Question 43

In terms of staring salary and tuition fee, how many schools are uniformly better (higher median starting salary AND lower tuition fee) than Dartmouth College?

A 1

B 2

C 3

D 4

VIEW SOLUTION

Instructions

Answer the questions based on the following information. Production pattern for number of units (in cubic feet) per day.

Days	1	2	3	4	5	6	7
Number of units	150	180	120	250	160	120	150

The finished goods are to be transported to the market by a truck having a capacity of 2000 cubic feet. Any finished goods (ready at the end of the day) retained overnight at the factory will incur a storage cost of Rs 5 per cubic foot for each night of storage. The hiring cost for the truck is Rs 1000 per day.

Question 44

If all the units should be sent to the market, then on which days should the trucks be hired to minimize the cost?

A 2nd, 4th, 6th, 7th

B 7th

C 2nd, 4th, 5th, 7th

D None of these

VIEW SOLUTION

Instructions

The table below presents the revenue (in million rupees) of four firms in three states. These firms, Honest Ltd., Aggressive Ltd., Truthful Ltd. and Profitable Ltd. are disguised in the table as A,B,C and D, in no particular order.

States	Firm A	Firm B	Firm C	Firm D
U.P	49	82	80	55
Bihar	69	72	70	65
M.P	72	63	72	65

Further, it is known that:

- In the state of MP, Truthful Ltd. has the highest market share.
- Aggressive Ltd.'s aggregate revenue differs from Honest Ltd.'s by Rs. 5 million.

Question 45

If Profitable Ltd.'s lowest revenue is from UP, then which of the following is true?

A Truthful Ltd.'s lowest revenues are from MP.

B Truthful Ltd.'s lowest revenues are from Bihar.

C Truthful Ltd.'s lowest revenues are from UP.

D No definite conclusion is possible.

VIEW SOLUTION

Data Interpretation for CAT Questions (download pdf)

Instructions

DIRECTIONS for the following three questions: In each question, there are two statements: A and B, either of which can be true or false on the basis of the information given below.

A research agency collected the following data regarding the admission process of a reputed management school in India.

Year	Gender	Number of Applications	Number of students who appeared for written test	Number of students who were called for interview	Number of students who were finally selected
2002	Male	61205	59981	684	171
	Female	19236	15389	138	48
2003	Male	63298	60133	637	115
	Female	45292	40763	399	84

Question 46

Statement A: The success rate of moving from written test to interview stage for males was worse than for females in 2003.

Statement B: The success rate of moving from written test to interview stage for females was better in 2002 than in 2003.

A Only Statement A is true.

B Only Statement B is true.

C Both the Statements are true.

D Neither of the two Statements is true.

VIEW SOLUTION

Instructions

Directions for the following two questions: Answer the questions based on the following information.

The batting average (BA) of a Test batsman is computed from runs scored and innings played — completed innings and incomplete innings (not out) in the following manner:

r_1 = Number of runs scored in completed innings

n_1 = Number of completed innings

r_2 = Number of runs scored in incomplete innings

n_2 = Number of incomplete innings

$BA = \frac{r_1+r_2}{n_1}$

To better assess a batsman's accomplishments, the ICC is considering two other measures MBA_1 and MBA_2 defined as follows:

$MBA_1 = \frac{r_1}{n_1} + \frac{n_2}{n_1} * max[0, (\frac{r_2}{n_2} - \frac{r_1}{n_1})]$

$MBA_2 = \frac{r_1+r_2}{n_1+n_2}$

Question 47

An experienced cricketer with no incomplete innings has BA of 50. The next time he bats, the innings is incomplete and he scores 45 runs. It can be inferred that

A BA and MBA_1 will both increase

B BA will increase and MBA_2 will decrease

C BA will increase and not enough data is available to assess change in MBA_1 and MBA_2

D None of these

VIEW SOLUTION

Question 48

Based on the above information which of the following is true?

A $MBA_1 \leq BA \leq MBA_2$

B $BA \leq MBA_2 \leq MBA_1$

C $MBA_2 \leq BA \leq MBA_1$

D None of these

VIEW SOLUTION

Instructions

Read the following information and answer the questions that follows: Ghosh Babu deposited a certain sum of money in a bank in 1986. The bank calculated interest on the principal at 10 percent simple interest, and credited it to the account once a year. After the 1st year, Ghosh Babu withdrew the entire interest and 20% of the initial amount. After the 2nd year, he withdrew the interest and 50% of the remaining amount. After the 3rd year, he withdrew the interest and 50% of the remaining amount. Finally after the 4th year, Ghosh Babu closed the account and collected the entire balance of Rs. 11,000.

Question 49

The year, at the end of which, Ghosh Babu withdrew the smallest amount was:

A First

B Second

C Third

D Fourth

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions based on the table given below.

The following table describes garments manufactured based upon the color and size for each lay. There are four sizes: M – medium, L – large, XL – extra large and XXL – extra extra large. There are three colors: yellow, red and white.

Lay	Number of Garments											
	Yellow				Red				White			
Lay No.	M	L	XL	XXL	M	L	XL	XXL	M	L	XL	XXL
1	14	14	7	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	42	42	21	0
3	20	20	10	0	18	18	9	0	0	0	0	0
4	20	20	10	0	0	0	0	0	30	30	15	0
5	0	0	0	0	24	24	12	0	30	30	15	0
6	22	22	11	0	24	24	12	0	32	32	16	0
7	0	24	24	12	0	0	0	0	0	0	0	0
8	0	20	20	10	0	2	2	1	0	0	0	0
9	0	20	20	10	0	0	0	0	0	22	22	11
10	0	0	0	0	0	26	26	13	0	20	20	10
11	0	22	22	11	0	26	26	13	0	22	22	11
12	0	0	2	2	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	20	20
14	0	0	0	0	0	0	0	0	0	0	22	22
15	0	0	10	10	0	0	2	2	0	0	22	22
16	0	0	0	0	1	0	0	0	1	0	0	0
17	0	0	0	0	0	5	0	0	0	0	0	0
18	0	0	0	0	0	32	0	0	0	0	0	0
19	0	0	0	0	0	32	0	0	0	0	0	0
20	0	0	0	0	0	5	0	0	0	0	0	0
21	0	0	0	18	0	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0	26	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	22
24	0	0	0	8	0	0	0	1	0	0	0	0
25	0	0	0	8	0	0	0	0	0	0	0	12
26	0	0	0	0	0	0	0	1	0	0	0	14
27	0	0	0	8	0	0	0	2	0	0	0	12
Production	76	162	136	97	67	194	89	59	135	198	195	156
Order	75	162	135	97	134	388	178	118	135	197	195	155
Surplus	1	0	1	0	0	0	0	0	0	1	0	1

Question 50

How many lays are used to produce XL yellow or XL white fabrics?

A 8

B 9

C 10

D 15

VIEW SOLUTION

Question 51

How many lays are used to produce XL fabrics?

A 15

B 16

C 17

D 18

VIEW SOLUTION

Question 52

How many lays are used to produce yellow fabrics?

A 10

B 11

C 12

D 14

VIEW SOLUTION

Instructions

The following table gives the quantities of apples (in tonnes) arriving at New Delhi market from various states in a particular year. The month in which demand was more than supply, the additional demand was met by the stock from cold storage.

Month	State			Cold Storage	Total
	HP	UP	J&K		
April	7	0	7	59	73
May	12	1	0	0	13
June	9,741	257	8,017	0	18,015
July	71,497	0	18,750	0	90,247
August	77,675	0	20,286	0	97,961
September	53,912	0	56,602	0	110,514
October	12,604	0	79,591	24	92,219
November	3,499	0	41,872	42	45,413
December	1,741	0	14,822	15	16,578
January	315	0	10,922	201	11,438
February	25	0	11,183	77	11,285
March	0	0	683	86	769

Question 53

If the yield per tree was 40 kg, then from how many trees were the apples supplied to New Delhi (in millions) during the year?

A 11.5

B 12.5

C 13.5

D Cannot be determined

VIEW SOLUTION

Question 54

Which state supplied the maximum number of apples?

A UP

B HP

C J & K

D Cold storage

VIEW SOLUTION

Instructions

The table below reports annual statistics related to rice production in select states of India for a particular year.

State	Total Area (in millions)	% of Area under rice cultivation	Production (in million tons)	Population (in millions)
Himachal Pradesh	6	20	1.2	6
Kerala	4	60	4.8	32
Rajasthan	34	20	6.8	56
Bihar	10	60	12	83
Karnataka	19	50	19	53
Haryana	4	80	19.2	21
West Bengal	9	80	21.6	80
Gujarat	20	60	24	51
Punjab	5	80	24	24
Madhya Prade3h	31	40	24.8	60
Tamilnadu	13	70	27.3	62
Maharashtra	31	50	48	97
Uttar Pradesh	24	70	67.2	166
Andhra Pradesh	28	80	112	76

Question 55

How many states have a per capita production of rice (defined as total rice production divided by its population) greater than Gujarat?

A 3

B 4

C 5

D 6

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

A study was conduced to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

	Country				
Rank	India	China	Japan	Malaysia	Thailand
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

Question 56

Three of the following four pairs of countries have identical levels of dissimilarity. Which pair is the odd one out?

A Malaysia and China

B China and Thailand

C Thailand and Japan

D Japan and Malaysia

VIEW SOLUTION

Instructions

These questions are based on the table below presenting data on percentage population covered by drinking water and sanitation facilities in selected Asian countries.

	Drinking water			Sanitation facilities		
	Urban	Rural	Total	Urban	Rural	Total
India	85	79	81	70	14	29
Bangladesh	99	96	97	79	44	48
China	97	56	67	74	7	24
Pakistan	82	69	74	77	22	47
Philippines	92	80	86	88	66	77
Indonesia	79	54	62	73	40	51
Sri Lanka	88	52	57	68	62	63
Nepal	88	60	63	58	12	1

(*Source:* World Resources 1998-99, p. 251, UNDP, UNEP and World Bank.)

Country A is said to dominate B or A > B if A has higher percentage in total coverage for both drinking water and sanitation facilities, and, B is said to be dominated by A, or B < A. A country is said to be on the coverage frontier if no other country dominates it. Similarly, a country is not on the coverage frontier if it is dominated by at least one other country.

Question 57

These relate to the above table with the additional provison that the gap between the population coverages of ‘sanitation facilities’ and ‘drinking water facilities’ is a measure of disparity in coverage.

The country with the least disparity in coverage of urban sector is

A India

B Pakistan

C Philippines

D None of these

VIEW SOLUTION

Instructions

Use the following information: Prakash has to decide whether or not to test a batch of 1000 widgets before sending them to the buyer.
In case he decides to test, he has two options:
a) Use test I ;
b) Use test II.

Test I cost Rs. 2 per widget. However, the test is not perfect. It sends 20% of the bad ones to the buyer as good. Test II costs Rs. 3 per widget. It brings out all the bad ones. A defective widget identified before sending can be corrected at a cost of Rs. 25 per widget. All defective widgets are identified at the buyer's end and penalty of Rs. 50 per defective widget has to be paid by Prakash.

Question 58

If there are 200 defective widgets in the lot, Prakash:

A may use either Test I or Test II

B should use Test I or not use any test

C should use Test II or not use any test.

D cannot decide.

VIEW SOLUTION

Instructions

DIRECTIONS for the following four questions:

A low-cost airline company connects ten India cities, A to J. The table below gives the distance between a pair of airports and the corresponding price charged by the company. Travel is permitted only from a departure airport to an arrival airport. The customers do not travel by a route where they have to stop at more than two intermediate airports.

Sector No.	Airport of Departure	Airport of Arrival	Distance between the airports	Price (Rs.)
1	A	B	560	670
2	A	C	790	1350
3	A	D	850	1250
4	A	E	1245	1600
5	A	F	1345	1700
6	A	G	1350	2450
7	A	H	1950	1850
8	B	C	1650	2000
9	B	H	1750	1900
10	B	I	2100	2450
11	B	J	2300	2275
12	C	D	460	450
13	C	F	410	430
14	C	G	910	1100
15	D	E	540	590
16	D	F	625	700
17	D	G	640	750
18	D	H	950	1250
19	D	J	1650	2450
20	E	F	1250	1700
21	E	G	970	1150
22	E	H	850	875
23	F	G	900	1050
24	F	I	875	950
25	F	J	970	1150
26	G	I	510	550
27	G	J	830	890
28	H	I	790	970
29	H	J	400	425
30	I	J	460	540

Question 59

If the prices include a margin of 10% over the total cost that the company incurs, what is the minimum cost per kilometer that the company incurs in flying from A to J?

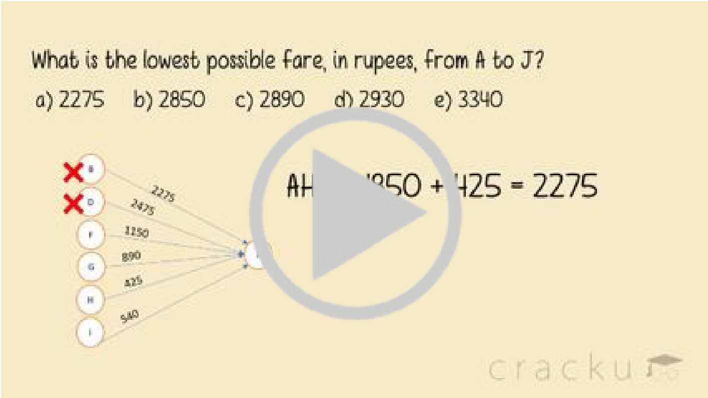
A 0.77

B 0.88

C 0.99

D 1.06

E 1.08



VIDEO SOLUTION

Instructions

XYZ organization got into the business of delivering groceries to home at the beginning of the last month. They have a two-day delivery promise. However, their deliveries are unreliable. An order booked on a particular day may be delivered the next day or the day after. If the order is not delivered at the end of two days, then the order is declared as lost at the end of the second day. XYZ then does not deliver the order, but informs the customer, marks the order as lost, returns the payment and pays a penalty for non-delivery. The following table provides details about the operations of XYZ for a week of the last month. The first column gives the date, the second gives the cumulative number of orders that were booked up to and including that day. The third column represents the number of orders delivered on that day. The last column gives the cumulative number of orders that were lost up to and including that day. It is known that the numbers of orders that were booked on the 11th, 12th, and 13th of the last month that took two days to deliver were 4, 6, and 8 respectively

Day	Cumilative orders booked	Orders delivered on day	Cumilative orders lost
13th	219	11	91
14th	249	27	92
15th	277	23	94
16th	302	11	106
17th	327	21	118
18th	332	13	120
19th	337	14	129

Question 60

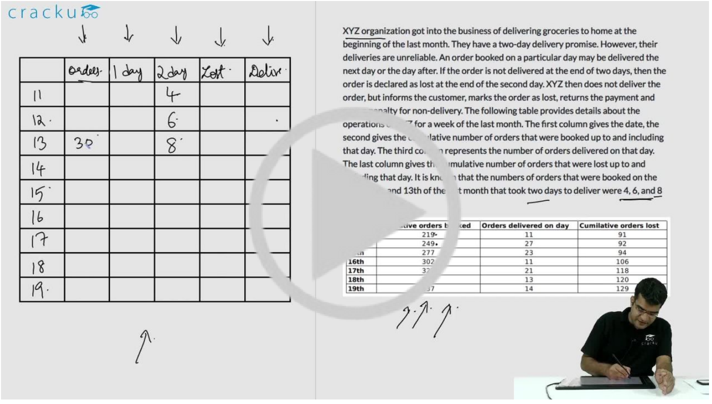
The delivery ratio for a given day is defined as the ratio of the number of orders booked on that day which are delivered on the next day to the number of orders booked on that day which are delivered on the second day after booking. On which of the following days, was the delivery ratio the highest?

A 15th

B 16th

C 13th

D 14th



VIDEO SOLUTION

Enroll for Excellent CAT/MBA courses

Instructions

In a certain board examination, students were to appear for examination in five subjects:
English, Hindi, Mathematics, Science and Social Science. Due to a certain emergency situation, a few of the examinations could not be conducted for some students. Hence, some students missed one examination and some others missed two examinations. Nobody missed more than two examinations.

The board adopted the following policy for awarding marks to students. If a student appeared in all five examinations, then the marks awarded in each of the examinations were on the basis of the scores obtained by them in those examinations.

If a student missed only one examination, then the marks awarded in that examination was the average of the best three among the four scores in the examinations they appeared for. If a student missed two examinations, then the marks awarded in each of these examinations was the average of the best two among the three scores in the examinations they appeared for. The marks obtained by six students in the examination are given in the table below. Each of them missed either one or two examinations.

	English	Hindi	Mathematics	Science	Social Science
Alva	80	75	70	75	60
Bithi	90	80	55	85	85
Carl	75	80	90	100	90
Deep	70	90	100	90	80
Esha	80	85	95	60	55
Foni	83	72	78	88	83

The following facts are also known.
I. Four of these students appeared in each of the English, Hindi, Science, and Social Science examinations.
II. The student who missed the Mathematics examination did not miss any other examination.
III. One of the students who missed the Hindi examination did not miss any other examination. The other student who missed the Hindi examination also missed the Science examination.

Question 61

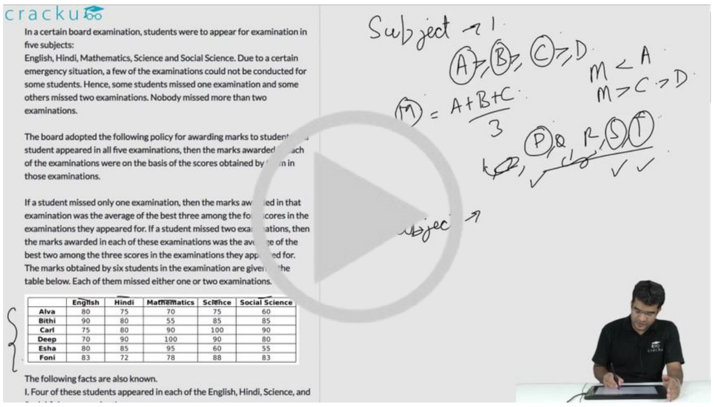
Who among the following did not appear for the Mathematics examination?

A Alva

B Carl

C Foni

D Esha



VIDEO SOLUTION

Instructions

Answer the questions based on the following information, which gives data about certain coffee producers in India.

	Production in '000 tonnes	Capacity Utilisation (%)	Sales ('000 tonnes)	Total Sales Value (Rs. in crores)
Brooke Bond	2.97	76.5	2.55	31.15
Nestle	2.48	71.2	2.03	26.75
Lipton	1.64	64.8	1.26	15.25
MAC	1.54	59.35	1.47	17.45
Total (Including Others)	11.6	61.3	10.67	132.8

Question 62

Among the four brands, the highest price for coffee per kilogram is for

A Nestle

B MAC

C Lipton

D Brooke Bond

VIEW SOLUTION

Instructions

In a certain board examination, students were to appear for examination in five subjects:
English, Hindi, Mathematics, Science and Social Science. Due to a certain emergency situation, a few of the examinations could not be conducted for some students. Hence, some students missed one examination and some others missed two examinations. Nobody missed more than two examinations.

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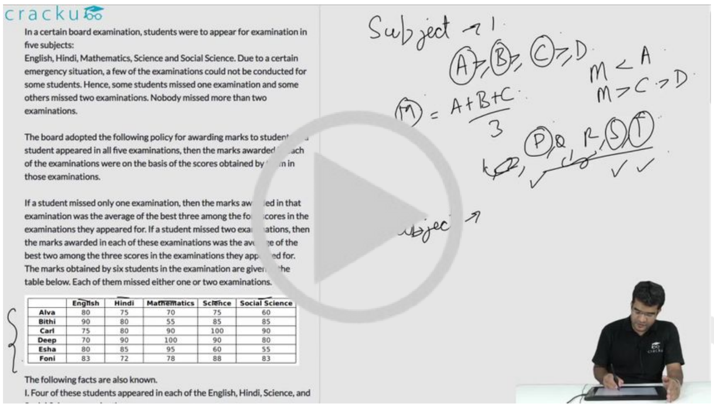
Carl	75	80	90	100	90
Deep	70	90	100	90	80
Esha	80	85	95	60	55
Foni	83	72	78	88	83

The following facts are also known.
I. Four of these students appeared in each of the English, Hindi, Science, and Social Science examinations.
II. The student who missed the Mathematics examination did not miss any other examination.
III. One of the students who missed the Hindi examination did not miss any other examination. The other student who missed the Hindi examination also missed the Science examination.

Question 63

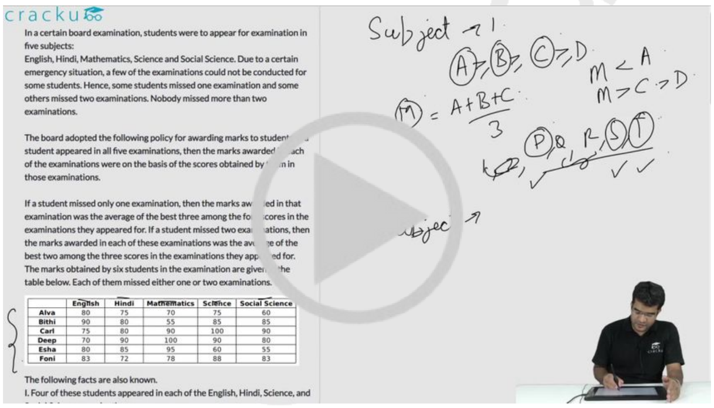
How many out of these six students missed exactly one examination?



VIDEO SOLUTION

Question 64

For how many students can we be definite about which examinations they missed?



VIDEO SOLUTION

Instructions

DIRECTIONS for the following four questions:

A low-cost airline company connects ten India cities, A to J. The table below gives the distance between a pair of airports and the corresponding price charged by the company. Travel is permitted only from a departure airport to an arrival airport. The customers do not travel by a route where they have to stop at more than two intermediate airports.

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5	A	F	1345	1700
6	A	G	1350	2450
7	A	H	1950	1850
8	B	C	1650	2000
9	B	H	1750	1900
10	B	I	2100	2450
11	B	J	2300	2275
12	C	D	460	450
13	C	F	410	430
14	C	G	910	1100
15	D	E	540	590
16	D	F	625	700
17	D	G	640	750
18	D	H	950	1250
19	D	J	1650	2450
20	E	F	1250	1700
21	E	G	970	1150
22	E	H	850	875
23	F	G	900	1050
24	F	I	875	950
25	F	J	970	1150
26	G	I	510	550
27	G	J	830	890
28	H	I	790	970
29	H	J	400	425
30	I	J	460	540

Question 65

What is the lowest possible fare, in rupees, from A to J?

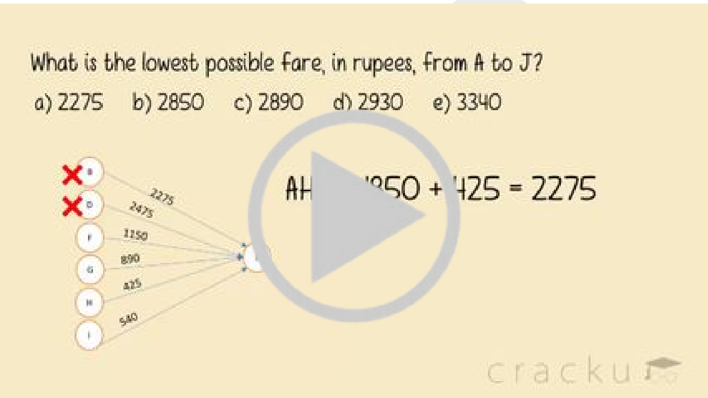
A 2275

B 2850

C 2890

D 2930

E 3340



VIDEO SOLUTION

Question 66

The company plans to introduce a direct flight between A and J. The market research results indicate that all its existing passengers travelling between A and J will use this direct flight if it is priced 5% below the minimum price that they pay at present. What should the company charge approximately, in rupees, for this direct flight?

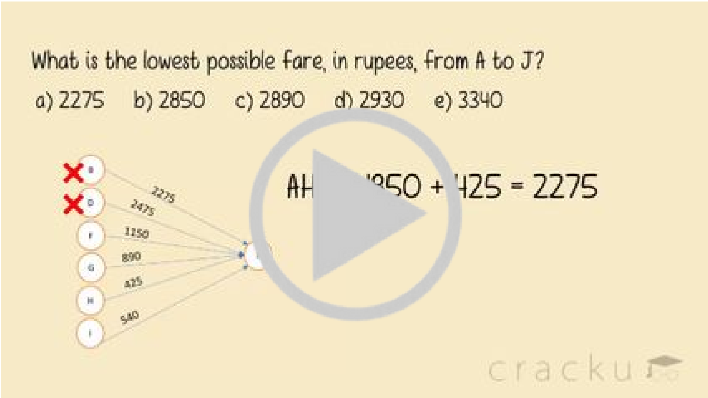
A 1991

B 2161

C 2707

D 2745

E 2783



VIDEO SOLUTION

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Question 67

If the airports C, D and H are closed down owing to security reasons, what would be the minimum price, in rupees, to be paid by a passenger travelling from A to J?

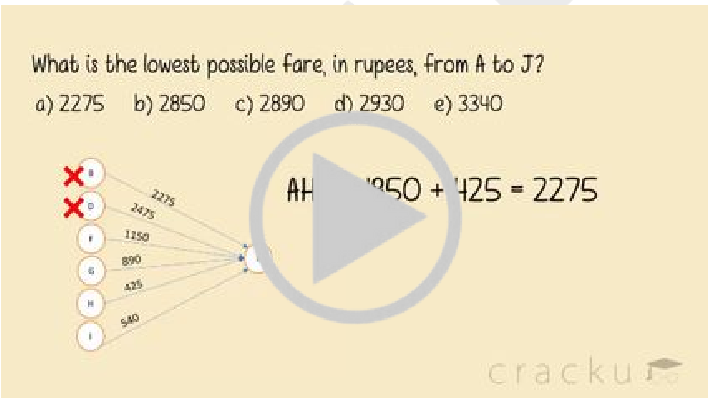
A 2275

B 2615

C 2850

D 2945

E 3190



VIDEO SOLUTION

Instructions

Directions for following three questions: Answer the following questions based on the information given below: There are 100 employees in an organization across five departments. The following table gives the department-wise distribution of average age, average basic pay and allowances. The gross pay of an employee is the sum of his/her basic pay and allowances.

There are limited numbers of employees considered for transfer/promotion across departments Whenever a person is transferred/promoted from a department of lower average age to a department of higher average age, he/she will get an additional allowance of 10% of basic pay over and above his/her current allowance. There will not be any change in pay structure if a person is transferred/promoted from a department with higher average age to a department with lower average age.

Department	Number of Employees	Average Age	Average Basic Pay	Allowances (% of Basic Pay)
HR	5	45	5000	70
Marketing	30	35	6000	80
Finance	20	30	6500	60
Business Development	35	42	7500	75
Maintenance	10	35	5500	50

Questions below are independent of each other.

Question 68

If two employees (each with a basic pay of Rs. 6000) are transferred from Maintenance department to HR department and one person (with a basic pay of Rs. 8000) was transferred from Marketing department to HR department, what will be the percentage change in average basic pay of HR department?

A 10.5%

B 12.5%

C 15%

D 30%

E 40%

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: In each question, there are two statements: A and B, either of which can be true or false on the basis of the information given below.

A research agency collected the following data regarding the admission process of a reputed management school in India.

Year	Gender	Number of Applications	Number of students who appeared for written test	Number of students who were called for interview	Number of students who were finally selected
2002	Male	61205	59981	684	171
	Female	19236	15389	138	48
2003	Male	63298	60133	637	115
	Female	45292	40763	399	84

Question 69

Statement A: In 2002, the number of females selected for the course as a proportion of the number of females who bought application forms, was higher than the corresponding proportion for males.

Statement B: In 2002, among those called for interview, males had a greater success rate than females.

A Only Statement A is true.

B Only Statement B is true.

C Both the Statements are true.

D Neither of the two Statements is true.

VIEW SOLUTION

Question 70

Statement A: The percentage of absentees in the written test among females decreased from 2002 to 2003.

Statement B: The percentage of absentees in the written test among males was larger than among females in 2003.

A Only Statement A is true.

B Only Statement B is true.

C Both the Statements are true.

D Neither of the two Statements is true.

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Details of the top 20 MBA schools in the US as ranked by US News and World Report, 1997 are given below.

School	Overall Ranking	Ranking by Academics	Ranking by Recruiters	Ranking by Placements	Median Starting Salary	% Employed	Annual Tuition Fee
Stanford University	1	1	3	1	$82,000	98.9	$23,100
Harvard University	2	1	2	4	$80,000	96.4	$23,840
University of Pennsylvania	3	1	4	2	$79,000	100.0	$24,956
Massachusetts Institute of Technology	4	1	4	3	$78,000	98.8	$23,900
University of Chicago	5	1	8	10	$65,000	98.4	$23,930
Northwestern University	6	1	1	11	$70,000	93.6	$23,025
Columbia University	7	9	10	5	$83,000	96.2	$23,830
Dartmouth College	8	12	11	6	$70,000	98.3	$23,700
Duke Univrsity	9	9	7	8	$67,500	98.5	$24,380
University of California—Berkeley	10	7	12	12	$70,000	93.7	$18,788
University of Virginia	11	12	9	9	$66,000	98.1	$19,627
University of Michigan—Ann Arbor	12	7	6	14	$65,000	99.1	$23,178
New York University	13	16	19	7	$70,583	97	$23,554
Carnegie Mellon University	14	12	18	13	$67,200	96.6	$22,200
Yale University	15	18	17	22	$65,000	91.5	$23,220
Univ. of North Carolina—Chapel Hill	16	16	16	16	$60,000	96.8	$14,333
University of California—Los Angeles	17	9	13	38	$65,000	82.2	$19,431
University of Texas—Austin	18	18	13	24	$60,000	97.3	$11,614
Indiana University—Bloomington	19	18	20	17	$61,500	95.2	$15,613
Cornell University	20	12	15	36	$64,000	85.1	$23,151

Question 71

How many schools in the list above have single digit rankings on at least 3 of the 4 parameters (overall ranking, ranking by academics, ranking by recruiters and ranking by placement)?

A 10

B 5

C 7

D 8

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

A study was conduced to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

	Country				
Rank	India	China	Japan	Malaysia	Thailand
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

Question 72

Which of the following countries is least dissimilar to India?

A China

B Japan

C Malaysia

D Thailand

VIEW SOLUTION

Instructions

In a Class X Board examination, ten papers are distributed over five Groups - PCB, Mathematics, Social Science, Vernacular and English. Each of the ten papers is evaluated out of 100. The final score of a student is calculated in the following manner. First, the Group Scores are obtained by averaging marks in the papers within the Group. The final score is the simple average of the Group Scores. The data for the top ten students are presented below. (Dipan's score in English Paper II has been intentionally removed in the table.)

Name of the Student	PCB Group			Mathematics Group	Social Science Group		Vernacular Group		English Group		Final Score
	Physics	Chemistry	Biology		History	Geography	Paper I	Paper II	Paper I	Paper II	
Ayesha (G)	98	96	97	98	95	93	94	96	96	98	96.2
Ram (B)	97	99	95	97	95	96	94	94	96	98	96.1
Dipan (B)	98	98	98	95	96	95	96	94	96	??	96
Sagnik (B)	97	98	99	96	96	98	94	97	92	94	95.9
Sanjiv (B)	95	96	97	98	97	96	92	93	95	96	95.7
Shreya (G)	96	89	85	100	97	98	94	95	96	95	95.5
Joseph (B)	90	94	98	100	94	97	90	92	94	95	95
Agni (B)	96	99	96	99	95	96	82	93	92	93	94.3
Pritam (B)	98	98	95	98	83	95	90	93	94	94	93.9
Tirna (G)	96	98	97	99	85	94	92	91	87	96	93.7

Note: B or G against the name of a student respectively indicates whether the student is a boy or a girl.

Question 73

How much did Dipan get in English Paper II?

A 94

B 96.5

C 97

D 98

E 99

VIEW SOLUTION

Question 74

Had Joseph, Agni, Pritam and Tirna each obtained Group Score of 100 in the Social Science Group,then their standing in decreasing order of final score would be;

A Pritam, Joseph, Tirna, Agni

B Joseph, Tirna, Agni, Pritam

C Pritam, Agni, Tirna, Joseph

D Joseph, Tirna, Pritam, Agni

E Pritam, Tirna, Agni, Joseph

VIEW SOLUTION

Instructions

The table below reports the gender, designation and age-group of the employees in an organization. It also provides information on their commitment to projects coming up in the months of January (Jan), February (Feb), March (Mar) and April (Apr), as well as their interest in attending workshops on: Business Opportunities (BO), Communication Skills (CS), and E-Governance (EG).

Serial No.	Name	Gender	Designation	Age-Group	Committed to Projects During	Interested in Workshop on
1	Anshul	M	Mgr	Y	Jan, Mar	CS, EG
2	Bushkant	M	Dir	I	Feb, Mar	BO, EG
3	Charu	F	Mgr	I	Jan, Feb	BO, CS
4	Dinesh	M	Exe	O	Jan, Apr	BO, CS, EG
5	Eashwaran	M	Dir	O	Feb, Apr	BO
6	Fatima	F	Mgr	Y	Jan, Mar	BO, CS
7	Gayatri	F	Exe	Y	Feb. Mar	EG
8	Hari	M	Mgr	I	Feb, Mar	BO, CS, EG
9	Indira	F	Dir	O	Feb. Apr	BO, EG
10	John	M	Dir	Y	Jan, Mar	BO
11	Kalindi	F	Exe	I	Jan, Apr	BO, CS, EG
12	Lavanya	F	Mgr	O	Feb. Apr	CS, EG
13	Mandeep	M	Mgr	O	Mar, Apr	BO, EG
14	Nandlal	M	Dir	I	Jan, Feb	BO, EG
15	Parul	F	Exe	Y	Feb, Apr	CS,EG
16	Rahul	M	Mgr	Y	Mar, Apr	CS, EG
17	Sunita	F	Dir	Y	Jan, Feb	BO, EG
18	Urvashi	F	Exe	I	Feb, Mar	EG
19	Yamini	F	Mgr	O	Mar, Apr	CS, EG
20	Zeena	F	Exe	Y	Jan, Mar	BO, CS, EG

M=Male, F= Female; Exe=Executive, Mgr=Manager, Dir=Director; Y=Young, I=In-between, O=Old

For each workshop, exactly four employees are to be sent, of which at least two should be Females and at least one should be Young. No employee can be sent to a workshop in which he/she is not interested in. An employee cannot attend the workshop on

- Communication Skills, if he/she is committed to internal projects in the month of January;
- Business Opportunities, if he/she is committed to internal projects in the month of February;
- E-governance, if he/she is committed to internal projects in the month of March.

Question 75

Assuming that Parul and Hari are attending the workshop on Communication Skills (CS), then which of the following employees can possibly attend the CS workshop?

A Rahul and Yamini

B Dinesh and Lavanya

C Anshul and Yamini

D Fatima and Zeena

VIEW SOLUTION

Instructions

The table below presents the revenue (in million rupees) of four firms in three states. These firms, Honest Ltd., Aggressive Ltd., Truthful Ltd. and Profitable Ltd. are disguised in the table as A,B,C and D, in no particular order.

States	Firm A	Firm B	Firm C	Firm D
U.P	49	82	80	55
Bihar	69	72	70	65
M.P	72	63	72	65

Further, it is known that:

- In the state of MP, Truthful Ltd. has the highest market share.

- Aggressive Ltd.'s aggregate revenue differs from Honest Ltd.'s by Rs. 5 million.

Question 76

What can be said regarding the following two statements?

Statement 1: Profitable Ltd. has the lowest share in MP market.

Statement 2: Honest Ltd.'s total revenue is more than Profitable Ltd.

A If Statement 1 is true then Statement 2 is necessarily true.

B If Statement 1 is true then Statement 2 is necessarily false.

C Both Statement 1 and Statement 2 are true.

D Neither Statement 1 nor Statement 2 is true.

VIEW SOLUTION

Question 77

What can be said regarding the following two statements?

Statement 1: Honest Ltd. has the highest share in the UP market.

Statement 2: Aggressive Ltd. has the highest share in the Bihar market.

A Both statements could be true.

B At least one of the statements must be true.

C At most one of the statements is true.

D None of the above.

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions based on the table given below.

The following table describes garments manufactured based upon the color and size for each lay. There are four sizes: M – medium, L – large, XL – extra large and XXL – extra extra large. There are three colors: yellow, red and white.

Lay	Number of Garments											
	Yellow				Red				White			
Lay No.	M	L	XL	XXL	M	L	XL	XXL	M	L	XL	XXL
1	14	14	7	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	42	42	21	0
3	20	20	10	0	18	18	9	0	0	0	0	0
4	20	20	10	0	0	0	0	0	30	30	15	0
5	0	0	0	0	24	24	12	0	30	30	15	0
6	22	22	11	0	24	24	12	0	32	32	16	0
7	0	24	24	12	0	0	0	0	0	0	0	0
8	0	20	20	10	0	2	2	1	0	0	0	0
9	0	20	20	10	0	0	0	0	0	22	22	11
10	0	0	0	0	0	26	26	13	0	20	20	10
11	0	22	22	11	0	26	26	13	0	22	22	11
12	0	0	2	2	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	20	20
14	0	0	0	0	0	0	0	0	0	0	22	22
15	0	0	10	10	0	0	2	2	0	0	22	22
16	0	0	0	0	1	0	0	0	1	0	0	0
17	0	0	0	0	0	5	0	0	0	0	0	0
18	0	0	0	0	0	32	0	0	0	0	0	0
19	0	0	0	0	0	32	0	0	0	0	0	0
20	0	0	0	0	0	5	0	0	0	0	0	0
21	0	0	0	18	0	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0	26	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	22
24	0	0	0	8	0	0	0	1	0	0	0	0
25	0	0	0	8	0	0	0	0	0	0	0	12
26	0	0	0	0	0	0	0	1	0	0	0	14
27	0	0	0	8	0	0	0	2	0	0	0	12
Production	76	162	136	97	67	194	89	59	135	198	195	156
Order	75	162	135	97	134	388	178	118	135	197	195	155
Surplus	1	0	1	0	0	0	0	0	0	1	0	1

Question 78

How many varieties of fabrics, which exceed the order, have been produced?

A 3

B 4

C 5

D 6

VIEW SOLUTION

CAT Previous Papers PDF

Instructions

Directions for the following four questions: Answer the questions based on the table given below concerning the busiest 20 international airports in the world.

No.	Name	International Airport Type	Code	Location	Passengers
1	Hartsfield	A	ATL	Atlanta, Georgia, USA	77,939,536
2	Chicago-O'Hare	A	ORD	Chicago, Illinois, USA	72,568,076
3	Los Angeles	A	LAX	Los Angeles, California, USA	63,876,561
4	Heathrow	E	LHR	London, United Kingdom	62,263,710
5	DFW	A	DFW	Dallas/Ft. Worth, Texas, USA	60,000,125
6	Haneda Airport	F	HND	Tokyo, Japan	54,338,212
7	Frankfurt Airport	E	FRA	Frankfurt, Germany	45,858,315
8	Roissy-Charles de Gaulle	E	CDG	Paris, France	43,596,943
9	San Francisco	A	SFO	San Francisco, California, USA	40,387,422
10	Denver	A	DIA	Denver, Colorado, USA	38,034,231
11	Amsterdam Schiphol	E	AMS	Amsterdam, Netherlands	36,781,015
12	Minneapolis - St. Paul	A	MSP	Minneapolis-St. Paul, USA	34,216,331
13	Detroit Metropolitan	A	DTW	Detroit, Michigan, USA	34,038,381
14	Miami	A	MIA	Miami, Florida, USA	33,899,246
15	Newark	A	EWR	Newark, New Jersey, USA	33,814,000
16	McCarran	A	LAS	Las Vegas, Nevada, USA	33,669,185
17	Phoenix Sky Harbor	A	PHX	Phoenix, Arizona, USA	33,533,353
18	Kimpo	FE	SEL	Seoul, Korea	33,371,074
19	George Bush	A	IAH	Houston, Texas, USA	33,089,333
20	John F. Kennedy	A	JFK	New York, New York, USA	32,003,000

Question 79

How many international airports of type 'A' account for more than 40 million passengers?

A 4

B 5

C 6

D 7

VIEW SOLUTION

Question 80

How many international airports not located in the USA handle more than 30 million passengers?

A 5

B 6

C 10

D 14

VIEW SOLUTION

Instructions

Directions for the next 5 questions: Answer these questions based on the data provided in the table below: Factory Sector by Type of Ownership. All figures in the table are in percent of the total for the corresponding column.

Sector	Factories	Employment	Fixed Capital	Gross Output	Value Added
Public	7	27.7	43.2	25.8	30.8
Central Govt.	1	10.5	17.5	12.7	14.1
State/local Govt.	5.2	16.2	24.3	11.6	14.9
Central & State/local Govt.	0.8	1	1.4	1.5	1.8
Joint	1.8	5.1	6.8	8.4	8.1
Wholly private	90.3	64.6	46.8	63.8	58.7
Others	0.9	2.6	3.2	2	2.4
Total	100	100	100	100	100

Question 81

A sector is considered 'pareto efficient' if its value added per employee and its value added per rupee of fixed capital is higher than those of all other sectors. Based on the table data, the pareto efficient sector is:

A Wholly private

B Joint

C Central and state/local

D Others

VIEW SOLUTION

Instructions

The following table provides data on the different countries and location of their capitals. (the data may not match the actual Latitude, Longitudes) Answer the following questions on the basis of this table.

S. No.	Country	Capital	Latitude	Longitude
1	Argentina	Buenos Aires	34.30 S	58.20 E
2	Australia	Canberra	35.15 S	149.08 E
3	Austria	Vienna	48.12 N	16.22 E
4	Bulgaria	Sofia	42.45 N	23.20 E
5	Brazil	Brasilia	15.47 S	47.55 E
6	Canada	Ottawa	45.27 N	75.42 E
7	Cambodia	Phnom Penh	11.33 N	104.55 E
8	Equador	Quito	0.15 S	78.35 E
9	Ghana	Accra	5.35 N	0.6 E
10	Iran	Teheran	35.44 N	51.30 E
11	Ireland	Dublin	53.20 N	6.18 E
12	Libya	Tripoli	32.49 N	13.07 E
13	Malaysia	Kuala Lumpur	3.9 N	101.41 E
14	Peru	Lima	12.05 S	77.0 E
15	Poland	Warsaw	52.13 N	21.0 E
16	New Zealand	Wellington	41.17 S	174.47 E
17	Saudi Arabia	Riyadh	24.41 N	46.42 E
18	Spain	Madrid	40.25 N	3.45 W
19	Sri Lanka	Colombo	6.56 N	79.58 E
20	Zambia	Lusaka	15.28 S	28.16 E

Question 82

What percentage of cities located within 10 E° and 40 E° (10° East and 40° East) lie in the Southern Hemisphere?

A 15%

B 20%

C 25%

D 30%

VIEW SOLUTION

Question 83

The number of cities whose names begin with a consonant and are in the Northern Hemisphere in the table

A exceeds the number of cities whose names begin with a consonant and are in the southern hemisphere by 1

B exceeds the number of cities whose names begin with a consonant and are in the southern hemisphere by 2

C is less than the number of cities whose names begin with a consonant and are in the east of the meridian by 1

D is less than the number of countries whose name begins with a consonant and are in the east of the meridian by 2

VIEW SOLUTION

Question 84

The ratio of the number of countries whose name starts with vowels and located in the southern hemisphere, to the number of countries, the name of whose capital cities starts with a vowel in the table above is

A 3 : 2

B 3 : 3

C 3 : 1

D 4 : 3

VIEW SOLUTION

Instructions

Read the following information and answer the questions that follows: Ghosh Babu deposited a certain sum of money in a bank in 1986. The bank calculated interest on the principal at 10 percent simple interest, and credited it to the account once a year. After the 1st year, Ghosh Babu withdrew the entire interest and 20% of the initial amount. After the 2nd year, he withdrew the interest and 50% of the remaining amount. After the 3rd year, he withdrew the interest and 50% of the remaining amount. Finally after the 4th year, Ghosh Babu closed the account and collected the entire balance of Rs. 11,000.

Question 85

The total interest, in rupees, collected by Ghosh Babu was:

A 12,000

B 20,000

C 4,000

D 11,000

VIEW SOLUTION

Instructions

Use the following information: Prakash has to decide whether or not to test a batch of 1000 widgets before sending them to the buyer.
In case he decides to test, he has two options:
a) Use test I ;
b) Use test II.

Test I cost Rs. 2 per widget. However, the test is not perfect. It sends 20% of the bad ones to the buyer as good. Test II costs Rs. 3 per widget. It brings out all the bad ones. A defective widget identified before sending can be corrected at a cost of Rs. 25 per widget. All defective widgets are identified at the buyer's end and penalty of Rs. 50 per defective widget has to be paid by Prakash.

Question 86

Prakash should not test if the number of bad widgets in the lot is:

A less than 100

B more than 200

C between 120 & 190

D Cannot be found out.

VIEW SOLUTION

Question 87

If there are 120 defective widgets in the lot, Prakash:

A should either use Test I or not test.

B should either use Test II or not test.

C should use Test I or Test II.

D should use Test I only.

VIEW SOLUTION

Question 88

If the number of defective widgets in the lot is between 200 and 400, Prakash:

A may use Test I or Test II

B should use Test I only.

C should use Test II only

D cannot decide.

VIEW SOLUTION

Instructions

Answer the questions based on the following information, which gives data about certain coffee producers in India.

	Production in '000 tonnes	Capacity Utilisation (%)	Sales ('000 tonnes)	Total Sales Value (Rs. in crores)
Brooke Bond	2.97	76.5	2.55	31.15
Nestle	2.48	71.2	2.03	26.75
Lipton	1.64	64.8	1.26	15.25
MAC	1.54	59.35	1.47	17.45
Total (Including Others)	11.6	61.3	10.67	132.8

Question 89

Which company out of the four companies mentioned above has the maximum unutilised capacity (in '000 tonnes)?

A Lipton

B Nestle

C Brooke Bond

D MAC

VIEW SOLUTION

Instructions

Answer the questions based on the following information. Production pattern for number of units (in cubic feet) per day.

Days	1	2	3	4	5	6	7
Number of units	150	180	120	250	160	120	150

The finished goods are to be transported to the market by a truck having a capacity of 2000 cubic feet. Any finished goods (ready at the end of the day) retained overnight at the factory will incur a storage cost of Rs 5 per cubic foot for each night of storage. The hiring cost for the truck is Rs 1000 per day.

Question 90

If the storage cost is reduced to Re 0.80 per cubic feet per day, then on which day(s), should the truck be hired?

A 4th

B 7th

C 4th and 7th

D None of these

VIEW SOLUTION

Instructions

The following table gives the quantities of apples (in tonnes) arriving at New Delhi market from various states in a particular year. The month in which demand was more than supply, the additional demand was met by the stock from cold storage.

Month	State			Cold Storage	Total
	HP	UP	J&K		
April	7	0	7	59	73
May	12	1	0	0	13
June	9,741	257	8,017	0	18,015
July	71,497	0	18,750	0	90,247
August	77,675	0	20,286	0	97,961
September	53,912	0	56,602	0	110,514
October	12,604	0	79,591	24	92,219
November	3,499	0	41,872	42	45,413
December	1,741	0	14,822	15	16,578
January	315	0	10,922	201	11,438
February	25	0	11,183	77	11,285
March	0	0	683	86	769

Question 91

Which state supplied the highest percentage of apples from the total apples supplied?

A HP

B UP

C J & K

D Cannot be determined

VIEW SOLUTION

Question 92

In which of the following periods was the supply greater than or equal to the demand?

A August-March

B June-October

C May-September

D Cannot be determined

VIEW SOLUTION

Question 93

Using the data in the earlier question, if there were 250 trees per hectare, then how many hectares of land was used?

A 9,400 hectares

B 49,900 hectares

C 50,000 hectares

D 49,450 hectares

VIEW SOLUTION

Instructions

Answer the questions based on the following information.

Krishna distributed 10-acre land to Gopal and Ram who paid him the total amount in the ratio 2 : 3. Gopal invested a further Rs. 2 lakh in the land and planted coconut and lemon trees in the ratio 5 : 1 on equal areas of land. There were a total of 100 lemon trees. The cost of one coconut was Rs. 5. The crop took 7 years to mature and when the crop was reaped in 1997, the total revenue generated was 25% of the total amount put in by Gopal and Ram together. The revenue generated from the coconut and lemon trees was in the ratio 3 : 2 and it was shared equally by Gopal and Ram as the initial amount spent by them were equal.

Question 94

What was the amount received by Gopal in 1997?

A Rs. 1.5 lakh

B Rs. 3 lakh

C Rs. 6 lakh

D None of these

VIEW SOLUTION

Question 95

What was the value of output per tree for coconuts?

A Rs. 36

B Rs. 360

C Rs. 3,600

D Rs. 240

VIEW SOLUTION

Question 96

What was the ratio of yields per acre of land for coconuts and lemons (in terms of number of lemons and coconuts)?

A 3 : 2

B 2 : 3

C 1 : 1

D Cannot be determined

VIEW SOLUTION

Instructions

These questions are based on the table below presenting data on percentage population covered by drinking water and sanitation facilities in selected Asian countries.

	Drinking water			Sanitation facilities		
	Urban	Rural	Total	Urban	Rural	Total
India	85	79	81	70	14	29
Bangladesh	99	96	97	79	44	48
China	97	56	67	74	7	24
Pakistan	82	69	74	77	22	47
Philippines	92	80	86	88	66	77
Indonesia	79	54	62	73	40	51
Sri Lanka	88	52	57	68	62	63
Nepal	88	60	63	58	12	1

(*Source:* World Resources 1998-99, p. 251, UNDP, UNEP and World Bank.)

Country A is said to dominate B or A > B if A has higher percentage in total coverage for both drinking water and sanitation facilities, and, B is said to be dominated by A, or B < A. A country is said to be on the coverage frontier if no other country dominates it. Similarly, a country is not on the coverage frontier if it is dominated by at least one other country.

Question 97

What are the countries on the coverage frontier?

A India and China

B Sri Lanka and Indonesia

C Philippines and Bangladesh

D Nepal and Pakistan

VIEW SOLUTION

Question 98

Which of the following statements are true?

A. India > Pakistan and India > Indonesia

B. India > China and India > Nepal

C. Sri Lanka > China

D. China > Nepal

A A and C

B B and D

C A, B and C

D B, C and D

VIEW SOLUTION

Question 99

Using only the data presented under Sanitation facilities columns, it can be concluded that rural population in India, as a percentage of its total population is approximately

A 76

B 70

C 73

D Cannot be determined

VIEW SOLUTION

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Question 100

India is not on the coverage frontier because

A. it is lower than Bangladesh in terms of coverage of drinking water facilities.

B. it is lower than Sri Lanka in terms of coverage of sanitation facilities.

C. it is lower than Pakistan in terms of coverage of sanitation facilities.

D. it is dominated by Indonesia.

A A and B

B A and C

C D

D None of these

VIEW SOLUTION

Question 101

These relate to the above table with the additional provison that the gap between the population coverages of 'sanitation facilities' and 'drinking water facilities' is a measure of disparity in coverage.

The country with the most disparity in coverage of rural sector is

A India

B Bangladesh

C Nepal

D None of these

VIEW SOLUTION

Instructions

The following table shows the top 20 busiest airports (in terms of passengers) in India. The Passengers column gives the number of passengers handled by each airport in a year.

No.	Name	Airport Rating	Code	Location	Passengers (in one year)
1	Rajiv Gandhi Airport	1	RGIA	Hyderabad, AP	7889987
2	Sardar Patel Airport	2	SVIA	Ahmedabad, Gujarat	7654321
3	Dibrugarh Airport	1	DGA	Dibrugarh, Assam	6890445
4	Gaya International Airport	1	GIA	Gaya, Bihar	6178901
5	Vskp International Airport	3	VIA	Vizag, AP	6000897
6	Shivaji Intnl Airport	1	CSIA	Mumbai, Maharashtra	5897561
7	Zero Airport	3	ZEA	Ziro, Arunachal Pradesh	4567892
8	Chandigarh Airport	1	CA	Chandigarh, Punjab	4237891
9	Vijayawada Airport	3	VZA	Vijayawada, AP	4145673
10	Purnea Int Airport	2	PIA	Purnea, Bihar	3898767
11	Indira Gandhi Airport	1	IGIA	New Delhi, Delhi	3587901
12	Hisar Airport	2	HA	Hisar, Haryana	3567879
13	Rajamundry Airport	1	RJA	Rajamundry, AP	3398764
14	Mehsana Airport	2	MA	Mehsana, Gujarat	3223786
15	Nagarjuna Sagar Airport	3	NSA	Karnool, AP	3217865
16	Bhuntar Airport	1	BIA	Kullu, Himachal	3134761
17	Kannur Intnl Airport	3	KIA	Kannur, Kerala	3113456
18	Sri Sathya Sai Airport	1	SSSA	Putaparthy, AP	3078934
19	Tirupati Intnl Airport	2	TIA	Tirupati, AP	2987434
20	HAL Airport	1	HBA	Bangalore, Karnataka	2923985

Question 102

How many airports outside of AP handle more than 40 lakh passengers each?

A 4

B 6

C 5

D 3

VIEW SOLUTION

Instructions

In a certain board examination, students were to appear for examination in five subjects:
English, Hindi, Mathematics, Science and Social Science. Due to a certain emergency situation, a few of the examinations could not be conducted for some students. Hence, some students missed one examination and some others missed two examinations. Nobody missed more than two examinations.

The board adopted the following policy for awarding marks to students. If a student appeared in all five examinations, then the marks awarded in each of the examinations were on the basis of the scores obtained by them in those examinations.

If a student missed only one examination, then the marks awarded in that examination was the average of the best three among the four scores in the examinations they appeared for. If a student missed two examinations, then the marks awarded in each of these examinations was the average of the best two among the three scores in the examinations they appeared for. The marks obtained by six students in the examination are given in the table below. Each of them missed either one or two examinations.

	English	Hindi	Mathematics	Science	Social Science
Alva	80	75	70	75	60
Bithi	90	80	55	85	85
Carl	75	80	90	100	90
Deep	70	90	100	90	80
Esha	80	85	95	60	55
Foni	83	72	78	88	83

The following facts are also known.
I. Four of these students appeared in each of the English, Hindi, Science, and Social Science examinations.
II. The student who missed the Mathematics examination did not miss any other examination.
III. One of the students who missed the Hindi examination did not miss any other examination. The other student who missed the Hindi examination also missed the Science examination.

Question 103

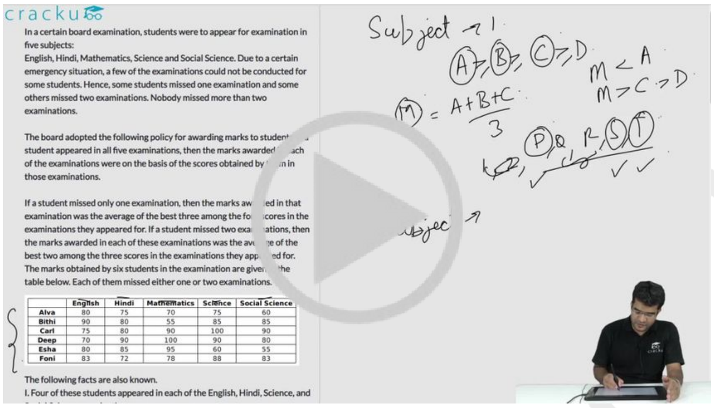
What BEST can be concluded about the students who missed the Science examination?

A Bithi and one out of Alva and Deep

B Alva and Bithi

C Deep and Bithi

D Alva and Deep



VIDEO SOLUTION

Answers

1.**D**	2.**D**	3.**B**	4.**D**	5.**C**	6.**B**	7.**B**	8.**A**
9.**B**	10.**A**	11.**C**	12.**A**	13.**B**	14.**A**	15.**A**	16.**A**
17.**C**	18.**A**	19.**B**	20.**D**	21.**A**	22.**6**	23.**C**	24.**D**
25.**C**	26.**C**	27.**B**	28.**D**	29.**A**	30.**B**	31.**B**	32.**A**
33.**C**	34.**C**	35.**D**	36.**A**	37.**C**	38.**D**	39.**C**	40.**D**
41.**A**	42.**E**	43.**B**	44.**C**	45.**C**	46.**D**	47.**B**	48.**D**
49.**D**	50.**D**	51.**A**	52.**D**	53.**B**	54.**C**	55.**B**	56.**D**
57.**C**	58.**A**	59.**B**	60.**D**	61.**B**	62.**A**	63.**3**	64.**4**
65.**A**	66.**B**	67.**C**	68.**B**	69.**D**	70.**A**	71.**D**	72.**A**
73.**C**	74.**A**	75.**A**	76.**B**	77.**C**	78.**B**	79.**B**	80.**B**
81.**C**	82.**B**	83.**D**	84.**A**	85.**A**	86.**A**	87.**D**	88.**C**
89.**D**	90.**C**	91.**C**	92.**C**	93.**D**	94.**A**	95.**B**	96.**D**
97.**C**	98.**B**	99.**C**	100.**D**	101.**A**	102.**B**	103.**A**	

Explanations

1.**D**

From table it is clear that for University of California - Berkeley annual tuition fee does not exceed $23,000 and median starting salary is at least $70,000. Hence option D.

VIEW SOLUTION

2.**D**

The number of people with age less than 40 are at least:

In no of children 0 category => 1 male and 1 female

In no of children 1 category => 1 male and 1 female

In no of children 2 category => 1 male and 1 female

In no of children 3 category => 2 male and 1 female

There are atleast 9 people who have age less than 40.

9 people out of 30 people => 30%

VIEW SOLUTION

3.**B**

Summation of sales value of all coffees (excluding others) = 31.15 + 26.75 + 15.25 + 17.45 = 90.6

Total value = 132.8

Others = 132.8 - 90.6 = 42.2

% share for others = $\frac{42.2}{132.8} \times 100$ = 31.77 $\approx$ 32

VIEW SOLUTION

4.**D**

From the given data it is clear that Dipan is the only student who obtained Group Scores of at least 95 in every group and obtained the highest Group Score in Social Science Group. Hence option D.

VIEW SOLUTION

5. **C**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B . Now if statement 1 is true then , Firm B is aggressive ltd. and Firm C is Honest Ltd. And also statement 2 is fulfilled.

VIEW SOLUTION

6. **B**

The common interest of all the people in option B is EG.

But, all the four are committed to projects in March => They cannot attend EG.

Hence, people in option 4 cannot attend a workshop.

VIEW SOLUTION

7. **B**

Gayatri and Urvashi are interested in only one workshop each, that is, EG. So they cannot attend more than one workshop.
Zeena is interested in all three workshops but she is committed to internal projects during the months of Jan and Mar. So she cannot attend the CS and EG workshops.
These are 3 executives gayatri , urvashi , zeena who can't attend more than 1 workshop.
Similarly using the same above logic we find that Anshul, Bushkant, Charu, Eshwaran, Fatima, Hari, Indiri, John, Mandeep, Nandlal, Rahul, Sunita and Yamini cannot also attend more than one workshop.
So, in total there are only 3 executives who cannot attent more than one work shop.

VIEW SOLUTION

8. **A**

Calculating the tons produced per hectare of rice cultivation for different states ,we get highest value for punjab and haryana , the ratio being above 5 for both . Hence option A.

VIEW SOLUTION

9. **B**

Value added per employee = Value added / Employment . So we can see that Central and State/local governments have highest Value added per employee = 1.8/1 = 1.8

VIEW SOLUTION

10. **A**

If the total number of factories is 100, then the total number of employees = 60 × 100 = 6000 of which 64.6% = 3876 work in wholly private factories. Since the number of wholly private factories = 90.3, So we have 3876/90.3 = 43. hence option A.

VIEW SOLUTION

11. **C**
Man's new gross income = 8000 + (8000*0.8) + (8000*0.1) = 15200

Average gross income of HR = 5000 + (5000*0.7) = 8500

New average = $\frac{8500*5+15200}{6}$ = 9616

% increase = $\frac{1116}{8500} * 100$ = 13.12% => 13% approximately.

VIEW SOLUTION

12. **A**
Percentage capacity of production for Lipton coffee = 64.8%
Let's say maximum production = x
So 1.64 = $x \times \frac{64.8}{100}$
or $x = 2.53$

VIEW SOLUTION

13. **B**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.
Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by coconuts is $3 * \frac{3}{3+2} = 3 * \frac{3}{5} = 1.8$ lakhs

Cost of a coconut is Rs. 5

Hence, the total output of coconuts is $\frac{180000}{5} = 36,000$

VIEW SOLUTION

14. **A**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by lemons is $3 * \frac{2}{3+2} = 3 * \frac{2}{5} = 1.2$ lakhs

Total land used to cultivate lemons is 5 acres.

So, revenue generated by the lemons per acre is $\frac{1.2}{5} = 0.24$ lakhs per acre

VIEW SOLUTION

15. **A**
For 160 defective pieces:
Using 1st test:
Cost of testing = $1000 \times 2 = 2000$
Cost correcting 80% pieces = $128 \times 25 = 3200$
Cost of penalty on 20% pieces = $32 \times 50 = 1600$

Total = 6800

Using 2nd test:
Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all pieces = $160 \times 25 = 4000$
Total = 7000

Using no test:
Cost of penalty on all pieces = $160 \times 50 = 8000$

Hence, he should use first test only.

VIEW SOLUTION

16. **A**
There are 6 airports from USA which are in top 10 busiest aiports. Hence 600/10 = 60%. Hence option A.

VIEW SOLUTION

17. **C**
Total passengers handled in top 5 airports are 336648008 . So percentage of passengers handled by heathrow airport is (62263710*100)/(336648008) which is approximately equal to 20%.

VIEW SOLUTION

18. **A**
This percentage is maximum in February in J&K; where the percentage is $\frac{11183}{11285} = 99.09$%

VIEW SOLUTION

19. **B**
Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Hence, Maximum withdrawal was after second year.

VIEW SOLUTION

20. **D**
Even if If the prices include a margin of 15% over the total cost that the company incurs, the total cost company incurs would be minimum for route AHJ i.e 2350 km. Hence option D.

VIDEO SOLUTION

21. **A**
Among the top ten students, only Dipan scored at least 95 in at least one paper from each of the groups. Hence option A.

VIEW SOLUTION

22. **6**
The airports with serial numbers 1, 3, 4, 6, 8 and 11 are of type '1' and handle more than 35 lakh passengers.

VIEW SOLUTION

23. **C**
Let's say the principal amount is P, Rate is R.
1st withdrawal = $PR + \frac{P}{5}$
Remaining money = $\frac{4P}{5}$

2nd withdrawal = $(\frac{4PR}{5} + \frac{2P}{5})$
Remaining money = $\frac{2P}{5}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$
Remaining money = $\frac{P}{5}$

4th withdrawal = $\frac{P}{5} + \frac{PR}{5}$ = 11000
Or P = 50000 (Putting R = 10%)

VIEW SOLUTION

24. **D**
Based on Condition **II**, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl missed his Mathematics examination.

Further, based on Condition **III**, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

$\therefore$ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option D: Esha and Foni.

VIDEO SOLUTION

25. **C**

There are atleast 4 respondents that fall into the 35 to 40 years age group (both inclusive) . So 400/30 = 13.333 % . Hence , the percentage of respondents that fall into the 35 to 40 years age group (both inclusive) is at least 13.333%.

VIEW SOLUTION

26. **C**

Let the age of an employee which were mutual transfer between Marketing and Finance departments be x(M to F) and y also age of one employee transfeered from Marketing to HR be z.

As a result, the average age of finance department increased by one year , so we have (30*20-x+y)/20 = 31 and that of Marketing department remained the same so we have (35*30+y-z-x)/29 = 35.

Solving both we have z = 15.

So new average age of HR department is (45*5 + 15)/6 = 40

Hence option C.

VIEW SOLUTION

27. **B**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option B: Alva and Deep.

VIDEO SOLUTION

28. **D**
10 lakh ton = 1 million ton. States whose annual rice production per million of population is at least 400,000 tons are Haryana , Gujarat, Punjab , Madhya pradesh, Tamil nadu, maharashtra, uttar pradesh,andhra pradesh . Hence 8 states.

VIEW SOLUTION

29. **A**
If A acquires firm B , the total sales of 2 firms is 50036 . now the percent market share can be given by 50036*100/89576 which is equal to around 55% . Hence option A.

VIEW SOLUTION

30. **B**
Maximum difference in ranks between Japan and India is 4 which is highest when compared to other countries. hence option B.

VIEW SOLUTION

31. **B**
We know that Compound productivity = Gross output / Fixed capital .So compound productivity for Public sector = 0.6, Central Government = 0.725, States/Local = 0.47, Central/States/Local = 1.07, Joint sector = 1.23 and wholly private = 1.36. Hence required is Wholly private, joint, central and state/local.

VIEW SOLUTION

32. **A**
The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

Date	Order Placed	1-day Delivery	2-day Delivery	Lost	Delivery done on the date
11			4		
12	14	7	6	1	
13	31	21	8	2	11
14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6=1=14.

Fraction of orders booked on 15th that were lost = 12/28

Fraction of orders booked on 16th that were lost = 2/25

Fraction of orders booked on 13th that were lost =2/31

Fraction of orders booked on 14th that were lost = 8/30.

∴ Option A is correct answer.

VIDEO SOLUTION

33. **C**
Let's total coffee production is = x

So $x \times \frac{61.3}{100} = 11.6$

x = 18.9 (nearly)

VIEW SOLUTION

34. **C**
The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

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15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

FRom the table we can determine that among options, number of orders booked on 13th are maximum.

Average time can be calculated as follows

	x	y	x+2y	x+y	(x+2y)/(x+y)
13	21	8	37	29	1.275862069
14	15	3	21	18	1.166666667
15	8	8	24	16	1.5
16	13	10	33	23	1.434782609

14 is the least

VIDEO SOLUTION

35. **D**

The maximum difference of the ranks between all traits is largest for Japan and Thailand which is 4. Hence option D.

VIEW SOLUTION

Quantitative Aptitude for CAT Questions (download pdf)

36. **A**

Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Maximum Interest was after the first year (i.e. PR)

VIEW SOLUTION

37. **C**

The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

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15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

The total number of orders placed on 13th = 21+8+2 = 31

FRom the table we can determine that among options, number of orders booked on 13th are maximum.

VIDEO SOLUTION

38. **D**

From the table we can see that , profitability in A , B ,C , D (4914)/(24568) , (4075)/(25468) , (4750)/(23752) , (3946)/(15782) . So clearly D has the highest profitability.

VIEW SOLUTION

39. **C**
There are atmost 23 people with age greater than 35 . Hence (23/30)*100 = 76.67 . So option C.

VIEW SOLUTION

40. **D**
We know that number of factories in joint sector is 1.8% = 2700, so number of factories in Central Government = 1% of (2700 x 100/1.8) = 1500.

Value added by Central Government = 14.1% of 14,000= Rs. 1974 crores.

So we get = (1974 crores)/(1500) = Rs. 1.32 crore

VIEW SOLUTION

41. **A**
Let total rural population in China be A and urban population in China be B.

Total population in India = T = A + B

Based on the sanitation facilities table,

$\frac{74}{100}$ B + $\frac{7}{100}$ A = $\frac{24}{100}$(A + B)

Solving, we get

$\frac{A}{B}$ = $\frac{50}{17}$

A : B = 50 : 17

Rural population % = $\frac{50}{50+17} \times 100$ = $\frac{50}{67} \times 100$ = 74.6%

Let total rural population in Indonesia be C and urban population in Indonesia be D.

Total population in India = T = C + D

Based on the sanitation facilities table,

$\frac{40}{100}$ C + $\frac{73}{100}$ D = $\frac{51}{100}$(C + D)

Solving, we get

$\frac{C}{D}$ = $\frac{22}{11}$

C : D = 2 : 1

Rural population % = $\frac{2}{2+1} \times 100$ = $\frac{2}{3} \times 100$ = 66.7%

Let total rural population in Philippines be R and urban population in Philippines be U.

Total population in India = T = R + U

Based on the sanitation facilities table,

$\frac{88}{100}$ U + $\frac{66}{100}$ R = $\frac{77}{100}$(U + R)

Solving, we get

$\frac{U}{R}$ = $\frac{11}{11}$

U : R = 1 : 1

Rural population % = $\frac{1}{1+1} \times 100$ = $\frac{1}{2} \times 100$ = 50%

Ascending order: Philippines, Indonesia, China

Therefore, answer is option A.

VIEW SOLUTION

42. **E**

Lets first consider ayesha's marks , her 7 marks will increase in geography . So increase in her average marks is 7/(2*5) = 0.7. So new average is 96.9 . </br>
Now for dipan , his 5 marks will increase in mathematics . So increase in his average 5/5 = 1 . So his new average is 96+1 = 97 which is greater than ayesha's . Other will clearly have a lower average than these both. Hence option E.

VIEW SOLUTION

43. **B**

There are only 2 colleges - New York university and stanford university which are uniformly better (higher median starting salary AND lower tuition fee) than Dartmouth College.

VIEW SOLUTION

44. **C**

The truck should be sent on a day on which maximum number of units can be sent. It happens on 2nd, 4th, 5th and 7th day. Hence the answer is b.

VIEW SOLUTION

45. **C**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

By Statement 1, Truthful is A or C.

If Profitable's lowest revenue are from UP, Profitable is either A or D.

By Statement 2, Honest and Aggressive are either A-D or B-C. As Profitable is either A or D, Honest and Aggressive will be one of B or C.

Hence, Truthful is A => Profitable is D.

Hence, Truthful's lowest revenues are from UP.

VIEW SOLUTION

46. **D**

The success rate of moving from written test to interview stage for males in 2003 can be given by 637/60133 and for females in the same year can be given by 399/40763 . We can see that the rate of males is clearly more that that of females. hence statement A is false.

Now the success rate of moving from written test to interview stage for females in 2002 was 138/15389 and for females in the year 2003 was 399/40763.So the rate in 2002 is less than that in 2003. . Hence both statements are false.

VIEW SOLUTION

47. **B**

With no incomplete innings and BA of 50 , lets assume $r_1 = 50$ and $n_1 = 1$.

Now we have $BA = MBA_2 = 50$

So we have $r_2 = 45$ and $n_2 = 1$.

We have $BA = 95$ and $MBA_2 = \frac{95}{2} < 50$.

Hence MBA_2 decreases.

So BA will increase and MBA_2 will decrease.

VIEW SOLUTION

48. **D**

Lets take $n_1 = n_2 = 10$ and $r_1 = r_2 = 100$.

Using given formula we have $BA = 20$ and $MBA_1 = MBA_2 = 10$.

So we have BA highest which is not the case in option A,B and C. Hence the correct option is D.

VIEW SOLUTION

49. **D**

Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Lowest withdrawal was after the fourth year.

VIEW SOLUTION

50. **D**

Manually counting lays which are are used to produce XL yellow or XL white fabrics. We get 15 number of such lays.

VIEW SOLUTION

51. **A**

Manually counting lays for which number of lays produced is non-zero . We get 15 lays. Hence option A.

VIEW SOLUTION

52. **D**

Manually counting non-zero value number of fabrics in yellow fabric category. We get 14. Hence option D.

1, 3, 4, 6, 7, 8, 9, 11, 12, 15, 21, 24, 25, 27

VIEW SOLUTION

53. **B**

To calculate the number of trees required, let us calculate the total number of tonnes supplied.

The total number of tonnes supplied equals 494,525.

One tree supplies 40 kilograms.

Hence, the number of trees required is $\frac{494525*1000}{40} \approx 12.5$ million trees

VIEW SOLUTION

54. **C**

From the table it is clear that the state which supplied the most number of apples is either HP or J&K.; UP and Cold Storage supplied less than a thousand apples in total.

The total number of apples supplied by HP is 231,028
The total number of apples supplied by J&K; is 262,735

Hence, the correct answer is J&K;

VIEW SOLUTION

55. **B**

Only 4 states Haryana , Punjab , Maharashtra, Andhra Pradesh have a per capita production of rice greater than Gujarat. Hence option B.

VIEW SOLUTION

56. **D**

The maximum difference in ranks in any of the parameters is termed to be dissimilarity. Difference between the ranks of Malaysia and China in Vision (V) = 5-1 =4 Difference between the ranks of China and Thailand in Vision (V) = 5-1 = 4 Difference between the ranks of Thailand and Japan in Decisiveness (D) = 5 -1 =4 The maximum rank difference between Japan and Malaysia is in Vision (V) and Negotiation skills (N). The maximum difference in rank is 4. Therefore, option D is the odd one.

VIEW SOLUTION

57. **C**

The least disparity => The least difference between the coverage in urban drinking water facilities and the coverage in urban sanitation facilities.

Phillipines has a disparity of 92-88 = 4, which is the least.

Hence, option C is the answer.

VIEW SOLUTION

58. **A**

When there are 200 defective pieces:
Using test 1
Cost of testing= 2000
Cost of correcting 80% pieces = 4000
Cost of penalty on 20% pieces = 2000
Total = 8000

Using test 2
Cost of testing = 3000
Cost of correcting all pieces = 5000

Total = 8000

Using no test
Cost of penalty on all pieces= 10000

Hence he can use either 1st test or second test.

VIEW SOLUTION

59. **B**
The minimum cost from A to J we know is 2275.

Let the CP to company be C

Since 10% over actual CP is the total price i.e. $\text{CP} \times 1.1 = 2275 \rightarrow CP = \frac{2275}{1.1}$

The total distance is 1950+1400=2350 Km.

Cost per Km = $\frac{\frac{2275}{1.1}}{2350}$ = Rs 0.88/Km

VIDEO SOLUTION

60. **D**
The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

Date	Order Placed	1-day Delivery	2-day Delivery	Lost	Delivery done on the date
11			4		
12	14	7	6	1	
13	31	21	8	2	11
14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

From the table we can determine that among options, number of orders booked on 13th are maximum.

For 15 the delivery ratio = 8/8 = 1

For 16 the delivery ratio = 13/10 = 1.3

For 13 the delivery ratio = 21/8 = 2.625

For 14 the delivery ratio = 15/3 = 5

Hence Option D

VIDEO SOLUTION

61. **B**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores.

For Alva: best 3 out of 4 - 80(English), 75(Hindi), 75(Science)

Avg. = 230/3 = 76.67 $\neq$ 70

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl most likely missed his Mathematics examination.

For Foni: best 3 out of 4 - 83(English), 83(Social Science), 88(Science)

Avg. = 254/3 = 84.67 $\neq$ 78

Hence, we observe that only Carl has missed his Mathematics examination. Hence, Option B is the correct answer.

VIDEO SOLUTION

62. **A**

The price per kilogram sales of each of the four brands is given below.

Brooke Bond - Rs. 122.16
Nestle - Rs. 131.77
Lipton - Rs. 121.03
MAC - Rs. 118.71

Hence, the correct answer is Nestle

VIEW SOLUTION

63. **3**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition **I**, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is **3**. {Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi)}

VIDEO SOLUTION

64. **4**

Based on Condition **II**, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition **III**, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

$\therefore$ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

$\therefore$ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

$\therefore$ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Except for Alva and Deep, we can conclusively comment of the missed subjects of the rest four. Hence, the correct answer is **4**.

VIDEO SOLUTION

65. **A**
From the table we can see that, the lowest price would be from A to H and H to J.

The cost of travel from A to H = Rs 1850

The cost of travel from H to J = Rs 425

Total cost = 1850 + 425 = Rs 2275.

VIDEO SOLUTION

66. **B**

From the table we can see that, the lowest price would be from A to H and H to J.

The cost of travel from A to H = Rs 1850

The cost of travel from H to J = Rs 425

Total cost = 1850 + 425 = Rs 2275

Lowest price = Rs 2275

95% of 2275 = Rs 2161

VIDEO SOLUTION

67. **C**

If the airports C, D and H are closed down the minimum price to be paid by a passenger travelling from A to J would be by first travelling to F and then from F to J.

The cost of travel from A to F = Rs 1700

The cost of travel from F to J = Rs 1150

Total cost = 1700 + 1150 = Rs 2850

VIDEO SOLUTION

68. **B**

The new average basic pay of HR dept. would be (5000*5 + 6000*2 + 8000)/8 = 45000/8.

So % change in basic pay is [(45000/8) - 5000]/5000 = 12.5%.

VIEW SOLUTION

69. **D**

In 2002, the number of females selected for the course as a proportion of the number of females who bought application forms was 48/19236 and for males was 171/61205. So rate for males was higher than that of female. Hence option A is false.

In 2002, among those called for interview, the success rate for females was 48/138 and for males 171/684. So the rate was higher for females .Both are false. Hence option D.

VIEW SOLUTION

70. **A**

The percentage of absentees in the written test among females in 2002 was 3847/19236 and in 2003 was 4529/45292. Thus percentage of absentees decreased from 02 to 03. Hence statement A is true.

The percentage of absentees in the written test among males in 2003 is 3065/63298 which is clearly less than that of female in 2003. Hence statement b is false . Hence option A.

VIEW SOLUTION

71. **D**

We can clearly make out from the given table that there are 8 schools in the list which have single digit rankings on at least 3 of the 4 parameters.

VIEW SOLUTION

72. **A**

If we find maximum of difference in ranks for all traits between India and others we find that the difference is least for India and china with just 2 points. Hence china is least dissimilar to India.

VIEW SOLUTION

73. **C**

Since we know the average , we can calculate the marks Dipan got in English Paper II (x) as ; 98+95+95.5+95+(96+x)/2 = 96*5 . We get 96+x = 96.5*2 . Thus x = 97 . Hence option C .

VIEW SOLUTION

74. **A**

Had Joseph, Agni, Pritam and Tirna each obtained Group Score of 100 in the Social Science Group , their final score would increase by 0.9, 0.9, 2.2, 2.1 respectively. Adding these to the final score then their standing in decreasing order of final score would be Pritam, Joseph, Tirna, Agni with scores of 96.1,95.9,95.8,95.2 respectively. Hence option A.

VIEW SOLUTION

75. **A**

Since we require atleast 2 women and 1 women is selected , other girl can be any one out of the yamini or lavanya. Now seeing the options and given data , we can infer that dinesh and anshul cannot attend as they have projects in january and rahul can attend it. So we have rahul and yamini. Hence option A.

VIEW SOLUTION

76. **B**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B .
Now if Profitable Ltd. has the lowest share in MP market then Firm B is profitable ltd. Honest ltd. would be any firm out of A and D. So now if statement 1 is true then statement 2 is false as honest will have more revenue than profitable. hence option c.

VIEW SOLUTION

77. **C**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B . Now if statement 1 is true then Honest ltd. would be firm B , and aggressive ltd. would be firm C. But now if we consider statement 2 we get Agressive ltd. as firm B and Honest ltd as firm C. Thus any 1 out of the 2 statements can be true and not both together. hence option C.

VIEW SOLUTION

78. **B**

Counting the number of variety of fabrics which have positive non-zero surplus, we get 4 such varieties. Hence option B.

VIEW SOLUTION

79. **B**

1 Million = 10 lakhs. Using the conversion and manually counting the international airports of type 'A' accounting for more than 40 million passengers, we get 5 such airports. Hence option B .

VIEW SOLUTION

80. **B**

All the top 20 busiest airports handle more than 30 million passengers. So, we have to just count the airports which are not located in the USA, out of these 20 airports. The airports ranked 4, 6, 7, 8, 11, 18 are located outside the USA. Thus, there are 6 such airports.
Hence, option B is the correct answer.

VIEW SOLUTION

81. **C**

We can find Value added/employment and value added/fixed capital for the different sectors. We get values as Wholly private 0.9 and 1.25; Joint sector 1.59 and 1.19; Central/State/Local 1.8, 1.28; others 0.92 and 0.75. Hence option C.

VIEW SOLUTION

82. **B**

There are 5 countries which lie from 10 Deg E to 40 Deg E , but out of those only 1 lie in southern hemisphere. hence 1*100/5 = 20%.

VIEW SOLUTION

83. **D**

There are 11 cities whose names begin with a consonant and are in the Northern Hemisphere in the table and there are 13 number of countries whose name begins with a consonant and are in the east of the meridian. Hence option D.

VIEW SOLUTION

84. **A**

Number of countries whose name starts with vowels and located in the southern hemisphere are 3 and number of countries, the name of whose capital cities starts with a vowel in the table is 2.

Hence the ratio 3:2.

VIEW SOLUTION

85. **A**

Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st Interest = PR

2nd Interest = $\frac{4PR}{5}$

3rd Interest = $\frac{2PR}{5}$

4th Interest = $\frac{PR}{5}$

Total interest = $\frac{12PR}{5} = 12000$

VIEW SOLUTION

86. **A**

We should consider three possible cases as 1. when he is using test 1
2. When he is using test 2
3. When he is using no test
Now we will choose a method where the total expenditure will be least.
So for option A, let's consider that number of defective pieces are 50.
Hence, while using test 1:
Cost of testing = $1000 \times 2 = 2000$
Correcting 80% of pieces = $40 \times 25 = 1000$
Penalty for 20% of pieces = $10 \times 50 = 500$
Total = 3500

For test 2:
Cost of testing = $1000 \times 3 = 3000$
Correcting all 50 pieces = $50 \times 25 = 1250$
Total= 4250

For no test:
Penalty for all 50 pieces = $50 \times 50 = 2500$

Hence cost is least when he is using no test while number of defective pieces is less than 100.

VIEW SOLUTION

87. **D**

When there are 120 widgets defective:

Using 1st test:
Cost of testing = $1000 \times 2 = 2000$
Cost of correcting 80% pieces = $96 \times 25 = 2400$
Cost of penalty on 20% pieces = $24 \times 50 = 1200$
Total cost = 5600

Using 2nd test:

Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all 120 pieces = $120 \times 25 = 3000$
Total cost = 6000

Using no test:

Total cost = 120 x 50 = 6000

Hence, he should use 1st test only.

VIEW SOLUTION

88. **C**
Let's say defective pieces are 300
Using test 1
Cost of testing= $1000 \times 2 = 2000$
Cost of correcting 80% pieces = $240 \times 25 = 6000$
Cost of penalty on 20% pieces = $60 \times 50 = 3000$
Total cost = 11000

Using test 2
Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all 300 pieces = $300 \times 25 = 7500$
Total cost = 10500

Using no test:
Cost of penalty on all pieces = $300 \times 50 = 15000$

Hence, he should use 2nd test.

VIEW SOLUTION

89. **D**
Maximum unutilized capacity => Minimum unitized capacity

MAC has the least utilized capacity.

VIEW SOLUTION

90. **C**
If the truck is hired only for the 7th day, the cost will be (150*6 + 180*5 + 120*4 + 250*3 + 160*2 + 120)*0.8 = 2776
So, total cost incurred = 2776 + 1000 = 3776

If a truck is hired on 4th day and 7th day, the cost incurred will be (150*3 + 180*2 + 120)*0.8 + 1000 + (160*2 + 120)*0.8 + 1000 = 3096

So, a truck should be hired on the 4th and 7th day.

VIEW SOLUTION

91. **C**
The answer to this question is the same as the state which supplied the maximum number of apples.

We can rule out UP and Cold Storage as they supplied significantly less than the other two.
Among the other two states,

The total number of apples supplied by HP is 231,028
The total number of apples supplied by J&K; is 262,735

Hence, the correct answer is J&K;

VIEW SOLUTION

92. **C**
The supply is greater than or equal to demand if the cold storage number is zero.

This is the case between the months May-September.

Hence, the correct option is option (c)

VIEW SOLUTION

93. **D**
In the earlier question, we determined that we needed 12.36 million trees per year.
Each hectare has 250 trees.

So, number of hectares of land used is $\frac{12.36*1000000}{250} \approx 49450$

VIEW SOLUTION

94. **A**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
As the revenue is split equally, Gopal's share in the revenue is $3 * \frac{1}{2} = 1.5$ lakhs

VIEW SOLUTION

95. **B**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by coconuts is $3 * \frac{3}{3+2} = 3 * \frac{3}{5} = 1.8$ lakhs

The number of coconut trees is 5 times the number of lemon trees = 100*5 = 500
So, the value of output per trees for coconuts is $\frac{180000}{500} = 360$

VIEW SOLUTION

96. **D**
We know the total revenue generated by coconuts and lemons respectively.
We also know the price of a coconut and can thus calculate the total number of coconuts produced per acre.
We don't know the price of a lemon and thus can't calculate the total number of lemons produced per acre.

So, the answer can't be determined.

VIEW SOLUTION

97. **C**
Bangladesh has the highest percentage in drinking water facilities. Let us see which countries it dominates. This is possible if it has greater sanitation facilities also as compared to the other country.

Bangladesh has better sanitation facilities than India, China, Pakistan and Nepal. So those Bangladesh dominates those four countries and they are not on the coverage frontier.

Similarly, Philippines has the highest percentage of sanitation facilities. Let us see which countries it dominates. This is possible if it has greater drinking water coverage also.

Philippines has greater drinking water coverage than all countries except Bangladesh. Hence, it dominates all the countries except Bangladesh. So, no other country is on the coverage frontier.

Hence, the only countries on the coverage frontier are Bangladesh and Philippines.

VIEW SOLUTION

98. **B**
India has lesser sanitation facilities than Pakistan. Hence it is false that India > Pakistan. So statement A is false.

India has greater sanitation facilities as well as drinking water coverage than China. Similar is the case between India and Nepal.
Hence, India > China and India > Nepal. So, statement B is true.

Sri Lanka has lesser drinking water coverage than China. Hence, statement C is false.

China has greater sanitation facilities as well as drinking water coverage than Nepal. Hence, China > Nepal and statement D is true.

So, the only two true statements are B and D.

VIEW SOLUTION

99. **C**
Let total rural population in India be R and urban population in India be U.

Total population in India = T = U + R

Based on the sanitation facilities table,

$\frac{70}{100}$ U + $\frac{14}{100}$ R = $\frac{29}{100}$(U + R)

Solving, we get

$\frac{U}{R}$ = $\frac{15}{41}$

U : R = 15 : 41

Rural population % = $\frac{41}{15+41} \times 100$ = $\frac{41}{56} \times 100$ = 73.2%

VIEW SOLUTION

100. **D**
For a country to be on the coverage frontier, it should not be dominated by any other country. That is no other country should have both (sanitation facilities and drinking water coverage) better than it.
The first three options, though true, are not the reason for India not being on the coverage frontier. That is because they don't mention the other parameter at all.

For example, option A doesn't mention sanitation facilites while options B and C don't mention drinking water coverage. Hence, none of A,B and C are the reason India is not on the coverage frontier.

Option D is factually wrong as Indonesia has lesser drinking water coverage than India and doesn't dominate it.

Hence, none of the options given are correct.

VIEW SOLUTION

101. **A**
The most disparity in rural sector is in India = 79-14 = 65

The disparity in all other countries is less than 65.

VIEW SOLUTION

102. **B**
The first 9 airports handle more than 40 lakh passengers each, out of which 3 are in AP and 6 are outside AP.

VIEW SOLUTION

103. **A**
Based on Condition **II**, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition **III**, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

<u>For Foni</u>: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option A: Bithi & one out of Alva and Deep.

VIDEO SOLUTION